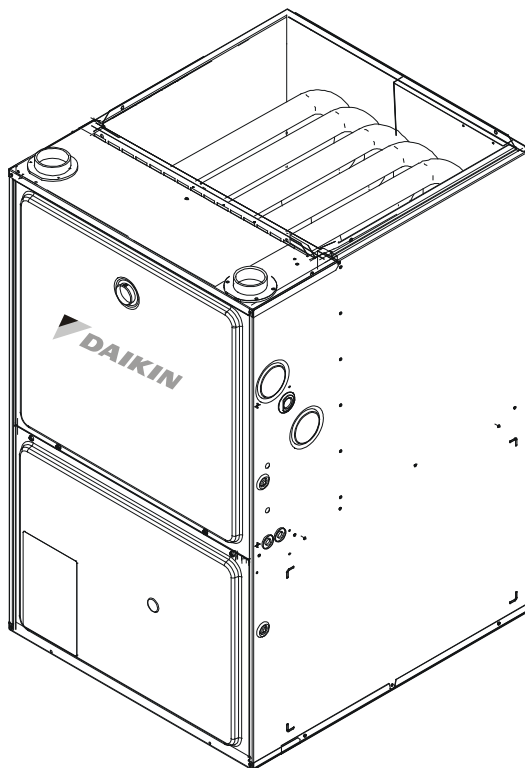


DAIKIN SERVICE INSTRUCTIONS

DM96SE AND DC96SE MODELS



WARNING

ONLY PERSONNEL THAT HAVE BEEN TRAINED TO INSTALL, ADJUST, SERVICE OR REPAIR (HEREINAFTER, "SERVICE") THE EQUIPMENT SPECIFIED IN THIS MANUAL SHOULD SERVICE THE EQUIPMENT. THE MANUFACTURER WILL NOT BE RESPONSIBLE FOR ANY INJURY OR PROPERTY DAMAGE ARISING FROM IMPROPER SERVICE OR SERVICE PROCEDURES. IF YOU SERVICE THIS UNIT, YOU ASSUME RESPONSIBILITY FOR ANY INJURY OR PROPERTY DAMAGE WHICH MAY RESULT. IN ADDITION, IN JURISDICTIONS THAT REQUIRE ONE OR MORE LICENSES TO SERVICE THE EQUIPMENT SPECIFIED IN THIS MANUAL, ONLY LICENSED PERSONNEL SHOULD SERVICE THE EQUIPMENT. IMPROPER INSTALLATION, ADJUSTMENT, SERVICING OR REPAIR OF THE EQUIPMENT SPECIFIED IN THIS MANUAL, OR ATTEMPTING TO INSTALL, ADJUST, SERVICE OR REPAIR THE EQUIPMENT SPECIFIED IN THIS MANUAL WITHOUT PROPER TRAINING MAY RESULT IN PRODUCT DAMAGE, PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.

PROP 65 WARNING FOR CALIFORNIA CONSUMERS



WARNING

Cancer and Reproductive Harm -
www.P65Warnings.ca.gov

0140M00517-A

This manual is to be used by qualified, professionally trained HVAC technicians only. Daikin does not assume any responsibility for property damage or personal injury due to improper service procedures or services performed by an unqualified person. The material in this manual does not supercede manufacturers installation and operation instructions.

Our continuing commitment to quality products may mean a change in specifications without notice.

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19001 Kermier Rd. Waller, Tx 77484

www.daikincomfort.com

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IMPORTANT INFORMATION

IMPORTANT INFORMATION

Pride and workmanship go into every product to provide our customers with quality products. It is possible, however, that during its lifetime a product may require service. Products should be serviced only by a qualified service technician who is familiar with the safety procedures required in the repair and who is equipped with the proper tools, parts, testing instruments and the appropriate service manual. **REVIEW ALL SERVICE INFORMATION IN THE APPROPRIATE SERVICE MANUAL BEFORE BEGINNING REPAIRS.**

IMPORTANT NOTICES FOR CONSUMERS AND SERVICERS

 **RECOGNIZE SAFETY SYMBOLS, WORDS AND LABELS**

**WARNING**

This unit should not be connected to, or used in conjunction with, any devices that are not design certified for use with this unit or have not been tested and approved by the manufacturer. Serious property damage or personal injury, reduced unit performance and/or hazardous conditions may result from the use of devices that have not been approved or certified by the manufacturer.

**WARNING**

HIGH VOLTAGE!
DISCONNECT ALL POWER BEFORE SERVICING OR INSTALLING THIS UNIT. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.



**WARNING**

TO PREVENT THE RISK OF PROPERTY DAMAGE, PERSONAL INJURY, OR DEATH, DO NOT STORE COMBUSTIBLE MATERIALS OR USE GASOLINE OR OTHER FLAMMABLE LIQUIDS OR VAPORS IN THE VICINITY OF THIS APPLIANCE.

IMPORTANT INFORMATION



WARNING

IF THE INFORMATION IN THESE INSTRUCTIONS IS NOT FOLLOWED EXACTLY, A FIRE OR EXPLOSION MAY RESULT CAUSING PROPERTY DAMAGE, PERSONAL INJURY OR LOSS OF LIFE.

- DO NOT STORE OR USE GASOLINE OR OTHER FLAMMABLE VAPORS AND LIQUIDS IN THE VICINITY OF THIS OR ANY OTHER APPLIANCE.
- WHAT TO DO IF YOU SMELL GAS:
 - DO NOT TRY TO LIGHT ANY APPLIANCE.
 - DO NOT TOUCH ANY ELECTRICAL SWITCH; DO NOT USE ANY PHONE IN YOUR BUILDING.
 - IMMEDIATELY CALL YOUR GAS SUPPLIER FROM A NEIGHBOR'S PHONE. FOLLOW THE GAS SUPPLIER'S INSTRUCTIONS.
 - IF YOU CANNOT REACH YOUR GAS SUPPLIER, CALL THE FIRE DEPARTMENT.
- INSTALLATION AND SERVICE MUST BE PERFORMED BY A QUALIFIED INSTALLER, SERVICE AGENCY OR THE GAS SUPPLIER.



WARNING

SHOULD OVERHEATING OCCUR OR THE GAS SUPPLY FAIL TO SHUT OFF, TURN OFF THE MANUAL GAS SHUTOFF VALVE EXTERNAL TO THE FURNACE BEFORE TURNING OFF THE ELECTRICAL SUPPLY.

To locate an authorized servicer, please consult your telephone book or the dealer from whom you purchased this product. For further assistance, please contact:

CONSUMER INFORMATION LINE - DAIKIN BRAND PRODUCTS

TOLL FREE 1-855-770-5678 (U.S. only)

email us at: customerservice@daikincomfort.com

fax us at: (713) 856-1821

(Not a technical assistance line for dealers.)

Outside the U.S., call 1-713-861-2500

(Not a technical assistance line for dealers.)

Your telephone company will bill you for the call.

IMPORTANT INFORMATION

FOR YOUR SAFETY READ BEFORE OPERATING

If you do not follow these instructions exactly, a fire or explosion may result causing property damage, personal injury or loss of life.

- A. This appliance does not have a pilot. It is equipped with an ignition device which automatically lights the burner. Do not try to light the burner by hand.
- B. BEFORE OPERATING smell all around the appliance area for gas. Be sure to smell next to the floor because some gas is heavier than air and will settle on the floor.
- WHAT TO DO IF YOU SMELL GAS**
- Do not try to light any appliance.
 - Do not touch any electric switch; do not use any phone in your building.
 - Immediately call your gas supplier from a neighbor's phone. Follow the gas supplier's instructions.
 - If you cannot reach your gas supplier, call the fire department.
- C. Use only your hand to push in or turn the gas control lever. Never use tools. If the lever will not push in or turn by hand, don't try to repair it, call a qualified service technician. Force or attempted repair may result in a fire or explosion.
- D. Do not use this appliance if any part has been underwater. Immediately call a qualified service technician to inspect the appliance and to replace any part of the control system and any gas control which has been underwater.

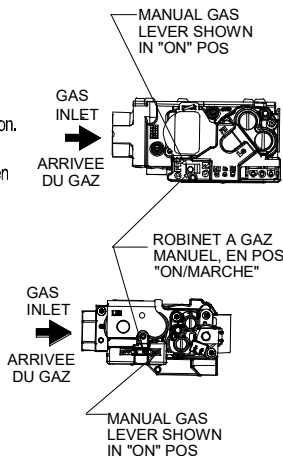
LIRE AVANT DE METTRE EN MARCHÉ

AVERTISSEMENT: Quiconque ne respecte pas à la lettre les instructions dans le présent manuel risque de déclencher un incendie ou une explosion entraînant des dommages matériels, des lésions corporelles ou la perte de vies humaines.

- A. Cet appareil ne comporte pas de veilleuse. Il est muni d'un dispositif d'allumage qui allume automatiquement le brûleur. Ne pas tenter d'allumer le brûleur manuellement.
- B. AVANT DE LE FAIRE FONCTIONNER, renifler tout autour de l'appareil pour détecter une odeur de gaz. Renifler près du plancher, car certains gaz sont plus lourds que l'air et peuvent s'accumuler au niveau du sol.
- QUE FAIRE S'IL Y A UNE ODEUR DE GAZ**
- Ne pas tenter d'allumer d'appareils.
 - Ne toucher à aucun interrupteur; ne pas vous servir des téléphones dans le bâtiment.
 - Appelez immédiatement votre fournisseur de gaz depuis un voisin. Suivez les instructions du fournisseur de gaz.
 - Si vous ne pouvez rejoindre le fournisseur de gaz, appelez le service des incendies.
- C. Ne pousser ou tourner la manette d'admission du gaz qu'à la main. Ne jamais employer d'outil à cette fin. Si la manette reste coincée, ne tenter pas de la réparer; appelez un technicien qualifié. Quiconque tente de forcer la manette ou de la réparer peut provoquer une explosion ou un incendie.
- D. Ne pas se servir de cet appareil s'il a été plongé dans l'eau, même partiellement. Faire inspecter l'appareil par un technicien qualifié et remplacer toute partie du système de contrôle et toute commande qui ont été plongés dans l'eau.

OPERATING INSTRUCTIONS

1. STOP! Read the safety information above on this label.
2. Set the thermostat to lowest setting.
3. Turn off all electric power to the appliance.
4. This appliance is equipped with an ignition device which automatically lights the burner. Do not try to light the burner by hand.
5. Push the gas control lever to "OFF" Position. Do not force.
6. Wait five (5) minutes to clear out any gas. Then smell for gas, including near the floor. If you then smell gas, STOP! Follow "B" in the safety information above on this label if you don't smell gas, go to next step.
7. Push gas control lever to "ON".
8. Replace access panel.
9. Turn on all electric power to the appliance.
10. Set thermostat to desired setting.
11. If the appliance will not operate, follow the instructions "To Turn Off Gas To Appliance" and call your service technician or gas company.

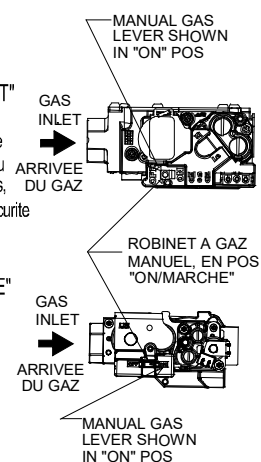


TO TURN OFF GAS TO APPLIANCE

1. Set the thermostat to lowest setting.
2. Turn off all electric power to the appliance if service is to be performed.
3. Push the gas control lever to "OFF" Position. Do not force.
4. Replace control access panel.

MISE EN MARCHÉ

1. ARRÊTEZ Lisez les instructions de sécurité dans la section supérieure de cette étiquette.
2. Régler le thermostat à la température la plus basse.
3. Couper l'alimentation électrique de l'appareil.
4. Cet appareil ménager étant doté d'un système d'allumage automatique, ne pas essayer d'allumer le brûleur manuellement.
5. Pousse le levier du contrôle du gaz à "OFF/ ARRÊT" position.
6. Attendre cinq (5) minutes pour laisser échapper tout le gaz. Renifler tout autour de l'appareil, y compris près du plancher, pour détecter une odeur de gaz. Si c'est le cas, ARRÊTEZ. Passer à l'étape B des instructions de sécurité sur la portion supérieure de cette étiquette. S'il n'y a pas d'odeur de gaz, passer à l'étape suivante.
7. Pousse le levier du contrôle du gaz à "ON/MARCHE" position.
8. Remettre en place le panneau d'accès.
9. Mettre l'appareil sous tension.
10. Régler le thermostat à la température désirée.
11. Si l'appareil ne se met pas en marche, suivre les instructions intitulées "Comment couper l'admission de gaz de l'appareil" et appeler un technicien qualifié ou le fournisseur de gaz.



POUR COUPER L'ADMISSION DE GAZ DE L'APPAREIL

1. Régler le thermostat à la température la plus basse.
2. Couper l'alimentation électrique de l'appareil s'il faut procéder à des opérations d'entretien.
3. Pousse le levier du contrôle du gaz à "OFF/ ARRÊT" position. Ne pas forcer.
4. Remettre en place le panneau d'accès.

PRODUCT IDENTIFICATION

The model and manufacturing number are used for positive identification of component parts used in manufacturing. Please use these numbers when requesting service or parts information.

D	M	9	6	S	E	0	6	0	3	B	N	A	A
1	2	3	4	5	6	7	8	9	10	11	12	13	14
Brand													
D - Daikin												Minor Revision	
												A - Initial Release	
												B - 1st Revision	
Configuration												Major Revision	
M - Upflow/Horizontal												A - Initial Release	
C - Downflow/Horizontal												B - 1st Revision	
K - Dedicated Upflow													
D - Dedicated Downflow													
AFUE												Nox	
97 - 97% AFUE												N - Natural Gas	
80 - 80% AFUE												X - Low NOx	
Gas Valve												Cabinet Width	
M - Modulating												A - 14"	
V - 2 Stage												B - 17.5"	
S - Single Stage												C - 21"	
												D - 24.5"	
Motor												Maximum CFM	
C - Variable Speed/ComfortNet												3 - 1200 CFM	
E - High Efficiency												4 - 1600 CFM	
S - Single Speed												5 - 2000 CFM	
MBTU/h													
40 - 40,000													
60 - 60,000													
80 - 80,000													
100 - 100,000													
120 - 120,000													

PRODUCT IDENTIFICATION

MODEL #	MFG. #	DESCRIPTION
DM96SE	DM96SE0403ANAA DM96SE0603BNAA DM96SE0803BNAA DM96SE0804CNAA DM96SE0805CNAA DM96SE1005CNAA DM96SE1205DNAA	Daikin® Brand 96% Single-Stage Furnace. 34.5" tall, up flow / horizontal installation, 120 volt silicon nitride hot surface ignition. Left or right gas pipe entry. Stainless steel tubular heat exchanger. Available cabinet widths are 14", 17.5", 21", & 24.5.

MODEL #	MFG. #	DESCRIPTION
DC96SE	DC96SE0403BNAA DC96SE0603BNAA DC96SE0804CNAA DM96SE1005CNAA DM96SE1205DNAA	Daikin® Brand 96% Single-Stage Furnace. 34.5" tall, up flow / horizontal installation, 120 volt silicon nitride hot surface ignition. Left or right gas pipe entry. Stainless steel tubular heat exchanger. Available cabinet widths are 14", 17.5", 21", & 24.5.

MODEL #	MFG #	DESCRIPTION
AFE18-60A	N/A	Fossil Fuel Kit. The AFE18-60A control is designed for use where the indoor coil is located above/downstream of a gas or fossil fuel furnace when used with a heat pump. It will operate with single and two stage heat pumps and single and two stage furnaces

MODEL #	MFG #	DESCRIPTION
DCVK-20 (CVENT - 2)	P1254001F	Concentric Vent Kits. For use with Daikin high efficiency furnaces. These kits are designed to allow terminations of a direct vent furnace to be "concentrically" vented through a wall or roof. These kits allow a single penetration to support terminations for both the vent/flue and the combustion air intake pipe. The DCVK-30 (2") and DCVK-30 (3") kits are certified for models listed above. See specification sheets on future models for use of the vent kits.
DCVK-30 (CVENT - 3)	P1254002F	
0170K00000S 0170K00001S	N/A	Side Wall Only Concentric Vent Kits. For use with Daikin high efficiency furnaces. These kits is to be used with 2" or 3" vent systems. The vent kit must terminate outside the structure. These kits are NOT intended for use with single pipe (indirect vent) installations.
EFR02	N/A	External Filter Rack Kit. This kit is intended to provide a location, external to the furnace casing, for installation of a permanent filter. The rack is mounted over the indoor air blower compartment area of either side panel, and provide filter retention as well as a location for attaching return air ductwork.
TK-400	TK-400	Furnace Twinning Kit. This kit allows furnaces to operate at the same time from a single thermostat. The two furnaces to be "twinned" must be the exact same model with their circulating air blowers set to deliver the same air flow at the same time. This kit cannot be used to control more than two furnaces.

PRODUCT IDENTIFICATION

MODEL #	MFG #	DESCRIPTION
LPLP03	N/A	<u>LP Gas Low Pressure Kit.</u> Designed for use with Daikin brand furnaces converted to LP gas. This kit includes harness adaptors to work with White-Rodgers single & two stage gas valves, Honeywell single and two-stage gas valves, as well as modulating gas valves.
LPM08	N/A	<u>LP Conversion Kit.</u> Installation of the LPM08 kit converts a 34.5" DM96SE/DC96SE for use with propane gas.
HASFK-4, 5, 6	N/A	<u>High Altitude Kits.</u> For use on Daikin brand 96% furnaces installed above 7,000 altitude. Kits contain orifices and pressure switches.

INSTALLATION CONSIDERATIONS

SAFETY

Please adhere to the following warnings and cautions when installing, adjusting, altering, servicing, or operating the furnace.

WARNING

TO PREVENT PERSONAL INJURY OR DEATH DUE TO IMPROPER INSTALLATION, ADJUSTMENT, ALTERATION, SERVICE OR MAINTENANCE, REFER TO THIS MANUAL. FOR ADDITIONAL ASSISTANCE OR INFORMATION, CONSULT A QUALIFIED INSTALLER, SERVICE AGENCY OR THE GAS SUPPLIER.

WARNING

TO PREVENT POSSIBLE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH DUE TO ELECTRICAL SHOCK, THE FURNACE MUST BE LOCATED TO PROTECT THE ELECTRICAL COMPONENTS FROM WATER.

CHARGE (ESD) PRECAUTIONS

NOTE: DISCHARGE BODY'S STATIC ELECTRICITY BEFORE TOUCHING UNIT. AN ELECTROSTATIC DISCHARGE CAN ADVERSELY AFFECT ELECTRICAL COMPONENTS.

Use the following precautions during furnace installation and servicing to protect the integrated control module from damage. By putting the furnace, the control, and the person at the same electrostatic potential, these steps will help avoid exposing the integrated control module to electrostatic discharge. This procedure is applicable to both installed and uninstalled (ungrounded) furnaces.

1. Disconnect all power to the furnace. Do not touch the integrated control module or any wire connected to the control prior to discharging your body's electrostatic charge to ground.
2. Firmly touch a clean, unpainted, metal surface of the furnace near the control. Any tools held in a person's hand during grounding will be discharged.
3. Service integrated control module or connecting wiring following the discharge process in Step 2. Use caution not to recharge your body with static electricity; (i.e., do not move or shuffle your feet, do not touch ungrounded objects, etc.). If you come in contact with an ungrounded object, repeat Step 2 before touching control or wires.
4. Discharge any static electricity from your body to ground before removing a new control from its container. Follow Steps 1 through 3 if installing the control on a furnace. Return any old or new controls to their containers before touching any ungrounded object.

WARNING

POSSIBLE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH DUE TO FIRE, EXPLOSION, SMOKE, SOOT, CONDENSATION, ELECTRICAL SHOCK OR CARBON MONOXIDE MAY RESULT FROM IMPROPER INSTALLATION, REPAIR, OPERATION, OR MAINTENANCE OF THIS PRODUCT.

WARNING

TO PREVENT PROPERTY DAMAGE, PERSONAL INJURY OR DEATH DUE TO FIRE, DO NOT INSTALL THIS FURNACE IN A MOBILE HOME, TRAILER, OR RECREATIONAL VEHICLE.

To ensure proper furnace operation, install, operate, maintain and service the furnace in accordance with the installation, operation and service instructions, all local building codes and ordinances. In their absence, follow the latest edition of the National Fuel Gas Code (NFPA 54/ANSI Z223.1), and/or CAN/CGA B149 Installation Codes, local plumbing or waste water codes, and other applicable codes.

A copy of the National Fuel Gas Code (NFPA 54/ANSI Z223.1) can be obtained from any of the following:

**American National Standards Institute
1430 Broadway
New York, NY 10018
National Fire Protection Association
1 Batterymarch Park
Quincy, MA 02269**

**CSA International
8501 East Pleasant Valley
Cleveland, OH 44131**

A copy of the CAN/CGA B149 Installation Codes can be obtained from:

**CSA International
178 Rexdale Boulevard
Etobicoke, Ontario, Canada M9W, 1R3**

THERMOSTAT REQUIREMENTS

DM96SE and DC96SE model furnaces require a single stage heating thermostat for operation. It is recommended that a high quality thermostat with a "C" terminal is used to operate the furnace.

GAS SUPPLY AND PIPING

The furnace rating plate includes the approved furnace gas input rating and gas types. The furnace must be equipped to operate on the type of gas applied. This includes any conversion kits required for alternate fuels and/or high altitude.



CAUTION

TO PREVENT UNRELIABLE OPERATION OR EQUIPMENT DAMAGE, THE INLET GAS SUPPLY PRESSURE MUST BE AS SPECIFIED ON THE UNIT RATING PLATE WITH ALL OTHER HOUSEHOLD GAS FIRED APPLIANCES OPERATING.

Inlet gas supply pressures must be maintained within the ranges specified below. The supply pressure must be constant and available with all other household gas fired appliances operating. The minimum gas supply pressure must be maintained to prevent unreliable ignition. The maximum must not be exceeded to prevent unit overfiring.

DM96SE MINIMUM CLEARANCES TO COMBUSTIBLE MATERIALS

POSITION	SIDES	REAR	FRONT	BOTTOM	FLUE	TOP
Upflow	0"	0"	1"	C	0"	1"
Horizontal	6"	0"	1"	C	0"	4"

C = Combustible: if placed on combustible floor, the floor MUST be wood ONLY.

For servicing or cleaning, a 24" front clearance is recommended.

Unit connections (electrical, flue, and drain) may necessitate great clearances than the minimum clearances listed above.

In all cases, accessibility clearance must take precedence over from the enclosure where accessibility clearances are greater.

Approved for line contact in the horizontal position.

DC96SE MINIMUM CLEARANCES TO COMBUSTIBLE MATERIALS

POSITION	SIDES	REAR	FRONT	BOTTOM	FLUE	TOP
Downflow	0"	0"	1"	NC	0"	1"
Horizontal	6"	0"	1"	C	0"	4"

C = Combustible: if placed on combustible floor, the floor MUST be wood ONLY.

NC = Non-Combustible: A combustible floor sub-base must be used for installation on combustible flooring.

Notes:

For servicing or cleaning, a 24" front clearance is recommended.

Unit connections (electrical, flue, and drain) may necessitate great clearances than the minimum clearances listed above.

In all cases, accessibility clearance must take precedence over from the enclosure where accessibility clearances are greater.

Our 34.5" DM96SE & DC96SE furnaces are part of our newly redesigned line of shorter chassis furnaces. It comes in two models: an up flow / horizontal model and a down flow / horizontal model.

The up flow / horizontal 34.5" furnace is available in the following models:

DM96SE0403ANAA
DM96SE0805CNAA
DM96SE0603BNAA
DM96SE0803BNAA
DM96SE0804CNAA
DM96SE1005CNAA
DM96SE1205DNAA

The down flow / horizontal 34.5" furnace is available in the following models:

DC96SE0403BNAA
DC96SE0603BNAA
DC96SE0804CNAA
DC96SE1005CNAA
DC96SE1205DNAA

Inlet gas supply pressures must be maintained within the ranges specified below. The supply pressure must be constant and available with all other household gas fired appliances operating. The minimum gas supply pressure must be maintained to prevent unreliable ignition. The maximum must not be exceeded to prevent unit overfiring.

INLET GAS SUPPLY PRESSURE		
Natural Gas	Minimum: 4.5" w.c.	Maximum: 10.0" w.c.
Propane Gas	Minimum: 11.0" w.c.	Maximum: 13.0" w.c.

Always use a backup wrench when working with piping connected to the gas valve to keep it from turning. The orientation of the gas valve on the manifold must be maintained as shipped from the factory.

PROPANE GAS CONVERSION

! WARNING
POSSIBLE PROPERTY DAMAGE, PERSONAL DAMAGE, PERSONAL INJURY OR DEATH MAY OCCUR IF THE CORRECT CONVERSION KITS ARE NOT INSTALLED. THE APPROPRIATE KITS MUST BE APPLIED TO INSURE SAFE AND PROPER FURNACE OPERATION. ALL CONVERSIONS MUST BE PERFORMED BY A QUALIFIED INSTALLER OR AGENCY.

The unit is configured for natural gas. The appropriate manufacturer's propane gas conversion kit must be applied for propane gas installations..

GAS PIPING CHECKS

Before placing unit in operation or after servicing, leak test the unit and gas connections.

! WARNING
TO AVOID THE POSSIBILITY OF EXPLOSION OR FIRE, NEVER USE A MATCH OR OPEN FLAME TO TEST FOR LEAKS.

Check for leaks using an approved chloride-free soap and water solution, an electronic combustible gas detector, or other approved testing methods.

NOTE: NEVER EXCEED SPECIFIED PRESSURES FOR TESTING. HIGHER PRESSURE MAY DAMAGE THE GAS VALVE AND CAUSE SUBSEQUENT OVERFIRING, RESULTING IN HEAT EXCHANGER FAILURE. DISCONNECT THIS UNIT AND SHUTOFF VALVE FROM THE GAS SUPPLY PIPING SYSTEM BEFORE PRESSURE TESTING THE SUPPLY PIPING SYSTEM WITH PRESSURES IN EXCESS OF 1/2 PSIG (3.48 kPa). ISOLATE THIS UNIT FROM THE GAS SUPPLY PIPING SYSTEM BY CLOSING ITS EXTERNAL MANUAL GAS SHUTOFF VALVE BEFORE PRESSURE TESTING SUPPLY PIPING SYSTEM WITH TEST PRESSURES EQUAL TO OR LESS THAN 1/2 PSIG (3.48 kPa).

PROPANE GAS TANKS AND PIPING

! WARNING
PROPANE GAS IS HEAVIER THAN AIR AND ANY LEAKING GAS CAN SETTLE IN ANY LOW AREAS OR CONFINED SPACES. TO PREVENT PROPERTY DAMAGE, PERSONAL INJURY, OR DEATH DUE TO FIRE OR EXPLOSION CAUSED BY A PROPANE GAS LEAK, INSTALL A GAS DETECTION WARNING DEVICE.

A gas detecting warning system is the only reliable way to detect a propane gas leak. Iron oxide (rust) can reduce the level of odorant in propane gas. Do not rely on your sense of smell. Contact a local propane gas supplier about installing a gas detecting warning system. If the presence of gas is suspected, please refer to the warning on this page.

All propane gas equipment must conform to the safety standards of the National Board of Fire Underwriters, NBFU Manual 58.

For satisfactory operation, propane gas pressure must be 10 inch WC at the furnace manifold with all gas appliances in operation.

! WARNING
IF THE GAS FURNACE IS INSTALLED IN A BASEMENT, AN EXCAVATED AREA OR A CONFINED SPACE, IT IS STRONGLY RECOMMENDED TO CONTACT A PROPANE SUPPLIER TO INSTALL A GAS DETECTING WARNING DEVICE IN CASE OF A GAS LEAK.
• SINCE PROPANE GAS IS HEAVIER THAN AIR, ANY LEAKING GAS CAN SETTLE IN ANY LOW AREAS OR CONFINED SPACES. • PROPANE GAS ODORANT MAY FADE, MAKING THE GAS UNDETECTABLE EXCEPT WITH A WARNING DEVICE.

! WARNING
AN UNDETECTED GAS LEAK WILL CREATE A DANGER OF EXPLOSION OR FIRE. IF THE PRESENCE OF GAS IS SUSPECTED, FOLLOW THE INSTRUCTIONS ON THE COVER OF THIS MANUAL. FAILURE TO DO SO COULD RESULT IN SERIOUS PERSONAL INJURY OR DEATH.

WARNING

IF THE INFORMATION IN THESE INSTRUCTIONS IS NOT FOLLOWED EXACTLY, A FIRE OR EXPLOSION MAY RESULT CAUSING PROPERTY DAMAGE, PERSONAL INJURY OR LOSS OF LIFE.

- Do NOT STORE OR USE GASOLINE OR OTHER FLAMMABLE VAPORS AND LIQUIDS IN THE VICINITY OF THIS OR ANY OTHER APPLIANCE.
- **WHAT TO DO IF YOU SMELL GAS:**
 - Do NOT TRY TO LIGHT ANY APPLIANCE.
 - Do NOT TOUCH ANY ELECTRICAL SWITCH; DO NOT USE ANY PHONE IN YOUR BUILDING.
 - IMMEDIATELY CALL YOUR GAS SUPPLIER FROM A NEIGHBOR'S PHONE. FOLLOW THE GAS SUPPLIER'S INSTRUCTIONS.
 - If YOU CANNOT REACH YOUR GAS SUPPLIER, CALL THE FIRE DEPARTMENT.
- INSTALLATION AND SERVICE MUST BE PERFORMED BY A QUALIFIED INSTALLER, SERVICE AGENCY OR THE GAS SUPPLIER.

ELECTRICAL CONNECTIONS

WARNING

TO AVOID THE RISK OF ELECTRICAL SHOCK, WIRING TO THE UNIT MUST BE PROPERLY POLARIZED AND GROUNDED.

WARNING

HIGH VOLTAGE

DISCONNECT ALL POWER BEFORE SERVICING OR INSTALLING THIS UNIT. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY, OR DEATH.



CAUTION

LABEL ALL WIRES PRIOR TO DISCONNECTION WHEN SERVICING CONTROLS. WIRING ERRORS CAN CAUSE IMPROPER AND DANGEROUS OPERATION. VERIFY PROPER OPERATION AFTER SERVICING.

WIRING HARNESS

The wiring harness is an integral part of this furnace. Field alteration to comply with electrical codes should not be required. Wires are color coded for identification purposes. Refer to the wiring diagram for wire routings. If any of the original wire as supplied with the furnace must be replaced, it must be replaced with wiring material having a temperature rating of at least 105° C. Any replacement wiring must be copper conductor.

115 VOLT LINE CONNECTIONS

Before proceeding with electrical connections, ensure that the supply voltage, frequency, and phase correspond to that specified on the unit rating plate. Power supply to the furnace must be N.E.C. Class 1, and must comply with all applicable codes. The furnace must be electrically grounded in accordance with local codes or, in their absence, with the latest edition of The National Electric Code, ANSI NFPA 70 and/or The Canadian Electric Code CSA C22.1.

Use a separate fused branch electrical circuit containing properly sized wire, and fuse or circuit breaker. The fuse or circuit breaker must be sized in accordance with the maximum overcurrent protection specified on the unit rating plate. An electrical disconnect must be provided at the furnace location.

NOTE: LINE POLARITY MUST BE OBSERVED WHEN MAKING FIELD CONNECTIONS.

WARNING

IN 90° FURNACE UPRIGHT UPFLOW INSTALLATIONS, THE DRAIN TRAP MUST BE MOUNTED ON THE OPPOSITE SIDE OF THE UNIT FROM THE JUNCTION BOX. THIS WILL REDUCE THE RISK OF WATER REACHING THE JUNCTION BOX IN THE EVENT OF A BLOCKED DRAIN CONDITION.

Connect hot, neutral, and ground wires as shown in the wiring diagram located on the unit's blower door. Line polarity must be observed when making field connections. Line voltage connections can be made through either the right or left side panel.

CAUTION

EDGES OF SHEET METAL HOLES MAY BE SHARP. USE GLOVES AS PRECAUTION WHEN REMOVING HOLE PLUGS.

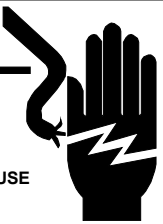
NOTE: WIRE ROUTING MUST NOT INTERFERE WITH CIRCULATOR BLOWER OPERATION, FILTER REMOVAL, OR ROUTINE MAINTENANCE.

WARNING

TO AVOID THE RISK OF ELECTRICAL SHOCK, INJURY, OR DEATH, THE FURNACE MUST BE ELECTRICALLY GROUNDED IN ACCORDANCE WITH LOCAL CODES OR, IN THEIR ABSENCE, WITH THE LATEST EDITION OF THE NATIONAL ELECTRIC CODE.

115 VOLT LINE CONNECTION OF ACCESSORIES (ELECTRONIC AIR CLEANER)


WARNING



HIGH VOLTAGE
DISCONNECT ALL POWER BEFORE SERVICING OR CHANGING ANY ELECTRICAL WIRING. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.

Accessory Load Specifications	
Electronic Air Cleaner	1.0 Amp maximum at 120 VAC

Turn OFF power to the furnace before installing any accessories. Follow the humidifier or air cleaner manufacturers' instructions for locating, mounting, grounding, and controlling these accessories.

24 VOLT HUMIDIFIER

The yellow wire connected to the I.D. Blower pressure switch is powered anytime the pressure switch is closed and provides 24 VAC humidifier control. Remove the yellow wire and connect the supplied brown jumper wire to the pressure switch terminal. Reconnect the yellow wire to the "piggyback" terminal on the brown jumper and then connect the 24 VAC line of the humidifier to the stripped end of the brown wire. Using a wire nut or a field-supplied quick connect terminal can make this connection. The wiring must conform to all local and national codes. Connect the COM side of the humidifier to the B/C terminal on the furnace control board (or to the COM side of the 24 VAC transformer). **DO NOT CONNECT 115V HUMIDIFIER TO THESE TERMINALS.**

If it is necessary for the installer to supply additional line voltage wiring to the inside of the furnace, the wiring must conform to all local codes, and have a minimum temperature rating of 105°C. All line voltage wire splices must be made inside the furnace junction box.

The integrated control module electronic air cleaner terminals (EAC) are energized with 115 volts whenever the circulator blower is energized.

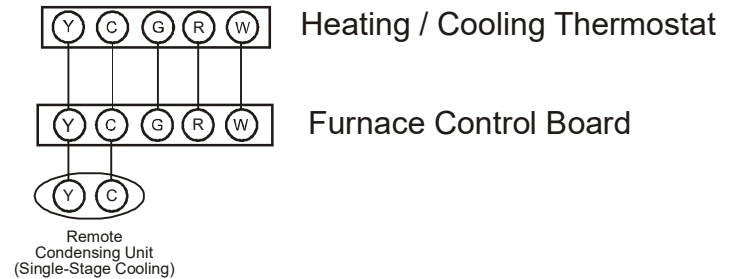
24 VOLT THERMOSTAT WIRING

NOTE: LOW VOLTAGE CONNECTIONS CAN BE MADE THROUGH EITHER THE RIGHT OR LEFT SIDE PANEL. WIRE ROUTING MUST NOT INTERFERE WITH CIRCULATOR BLOWER OPERATION, FILTER REMOVAL, OR ROUTINE MAINTENANCE.

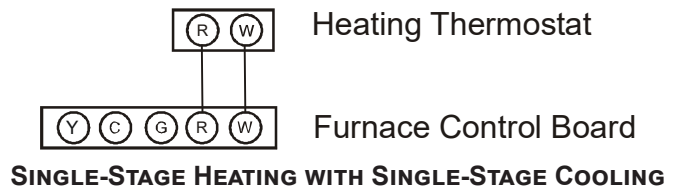
A 40 V.A. transformer and an integrated electronic control are built into the furnace to allow use with most cooling equipment. Consult the wiring diagram, located in this manual or on the blower door for further details of 115 Volt and 24 Volt wiring.

THERMOSTAT WIRING

The S model furnaces will have a "W" terminal and will use a single stage thermostat. The following drawing illustrates the typical field wiring for a heat only single stage system and a single stage heating/single stage cooling system. Refer to the following figures for proper connections to the integrated control module.



SINGLE-STAGE HEATING ONLY



CONTINUOUS FAN OPERATION

The S model furnace control will energize the low heat circulator fan speed when the fan switch on the thermostat is turned to the "ON" position.

CIRCULATOR BLOWER SPEED ADJUSTMENT


WARNING



HIGH VOLTAGE
DISCONNECT ALL POWER BEFORE CHANGING SPEED TAPS. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.

All furnaces ship at high speed for cooling and the speeds listed in the Specification Sheet for heating. These speeds should be adjusted by the installer to match the job requirements. See Specification Sheet for details.

Connect the correct motor leads to the COOL, HI HEAT, LOW HEAT and PARK terminals. If the heating speed equals cooling speed, use the jumper wire provided. All unused motor leads that are not connected to "PARK" terminals on the control must be taped to prevent shorts.

CIRCULATING AIR AND FILTERS

DUCTWORK - AIR FLOW

Duct systems and register sizes must be properly designed for the C.F.M. and external static pressure rating of the furnace. Ductwork should be designed in accordance with the recommended methods of "Air Conditioning Contractors of America" manual D.

A duct system should be installed in accordance with Standards of the National Board of Fire Underwriters for the Installation of Air Conditioning, Warm Air Heating and Ventilating Systems, Pamphlets No. 90A and 90B.

A return air filter is not supplied with the furnace. The installer must supply a means of filtering all of the return air. Filter(s) shall comply with UL900 or CAN/ULC-S111 Standards. If the furnace is installed without filters, the warranty will be voided.

Upflow furnaces with air delivery of less than 1800 CFM:
Use one side return or one bottom return ductwork connection.

Upflow furnaces with air delivery of 1800 CFM or higher:
Use two side returns or one side return and one bottom return connection.

Guide dimples locate the side and bottom return cutout locations. Use a straight edge to scribe lines connecting the dimples. Cut out the opening on these lines. An undersized opening will cause reduced airflow. For bottom return connection, remove the bottom of the cabinet before setting the furnace on the raised platform or return air duct.

A closed return duct system must be used, with the return duct connected to the furnace. **NOTE: Ductwork must never be attached to the back of the furnace.** Supply and return connections to the furnace may be made with flexible joints to reduce noise transmission, if desired. If a central return is used, a connecting duct must be installed between the unit and the utility room wall so the blower will not interfere with combustion air or draft. The room, closet, or alcove must not be used as a return air chamber.

When the furnace is used in connection with a cooling unit, the furnace should be installed in parallel with or on the upstream side of the cooling unit to avoid condensation in the heating element. With a parallel flow arrangement, the dampers or other means used to control the flow of air must be adequate to prevent chilled air from entering the furnace and, if manually operated, must be equipped with means to prevent operation of either unit unless the damper is in the full heat or cool position.

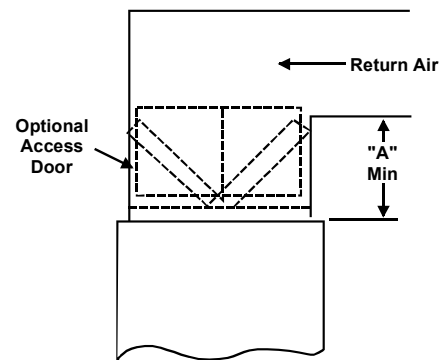
When the furnace is heating, the temperature of the return air entering the furnace must be between **55°F** and **100°F**.

UPRIGHT FILTER INSTALLATIONS

Depending on the installation and/or customer preference, differing filter arrangements can be applied. Filters can be installed in the central return register and a side panel external filter rack kit (upflow filter kit # EFR02). As an alternative a media air filter or electronic air cleaner can be used as the requested filter. Refer to the following minimum filter requirement charts for determination of the minimum filter area to ensure proper unit performance. The following figures show possible filter locations. **NOTE:** A ductwork access door must be used in counterflow applications to allow filter removal. If the filter rack is used, the side of the plenum must be at least as tall as dimension "A" shown in the following illustration. For dimension of "A" refer to the following chart.

COUNTERFLOW FILTER INSTALLATION

This furnace has provisions for the installation of return air filters at the counterflow top return. The furnace will accommodate the following filter sizes depending on cabinet size:



HORIZONTAL INSTALLATIONS

Filter(s) must be installed external to the furnace casing for horizontal installations. For most installations it is preferable to use a central return with filters installed in the duct behind the return air grill. In this way filter replacement is relatively simple by merely removing the grille, rather than going into the attic or crawl space.

The electronic air cleaner and media air cleaner are multi-positional high efficiency air filtration devices that can be installed in any position, except with the access door facing down. The best location for the air cleaner is in the return air duct next to the blower compartment. Before installing the air cleaner, consider the application. The electronic air cleaner must be readily accessible for periodic inspection and cleaning of the pre-filters and electronic cells while the media air cleaner must be readily accessible for periodic inspection and replacement of the Media Air Cleaner (AMU and GMU) filter cartridge (3 per carton) M0-1056, M1-1056, M2-1056, M8-1056, to maintain maximum efficiency and trouble-free operation. Carbon Filters 1156-3 and 1856-3 (set of 3) are also available. See Product Catalog for exact filter for your model.

SYSTEM OPERATION

POWER UP

The normal power up sequence is as follows:

1. 115 VAC power applied to furnace.
2. Integrated control module performs internal checks.
3. Integrated control module LED will light.
4. Integrated control module monitors safety circuits continuously.
5. Furnace awaits call from thermostat.

NORMAL HEATING SEQUENCE

1. R and W thermostat contacts close, initiating a call for heat.
2. Integrated control module performs safety circuit checks.
3. The induced draft blower is energized causing pressure switch contacts to close.
4. Igniter warm up begins after pre-purge is completed. The ignition control has a WARM UP PERIOD OF 17 SECONDS.
5. Gas valve opens at end of igniter warm up period, delivering gas to burners to establish flame.
6. The control checks for a signal from the flame sensor within seven (4) seconds after the gas valve is energized. Gas will continue to flow if a flame signal is present.
7. Circulator blower is energized on high heat speed following a fixed thirty (30) second blower on delay.
8. Furnace runs; integrated control module monitors safety circuits continuously.
9. R and W thermostat contacts open, allowing the gas valve to cycle off.
10. Induced draft blower is de-energized following a fifteen (15) second post purge.
11. Circulator blower is de-energized following a heat off delay period
12. Furnace awaits the next call from thermostat.

COOLING MODE

1. R and Y thermostat contacts close, initiating a call for cool.
2. Integrated control module performs safety circuit checks.
3. Outdoor fan and compressor are energized.
4. Circulator blower is energized on cool speed following a fixed five second on delay.
5. Furnace circulator blower and outdoor cooling unit run, integrated control module monitors safety circuits continuously.
6. R and Y thermostat contact open, completing the call for cool.
7. Outdoor fan and compressor are de-energized.
8. Circulator blower is de-energized following a fixed forty five (45) second cool off delay period.
9. Furnace awaits the next call from thermostat.

FAN ONLY MODE

The normal operational sequence in fan only mode is as follows:

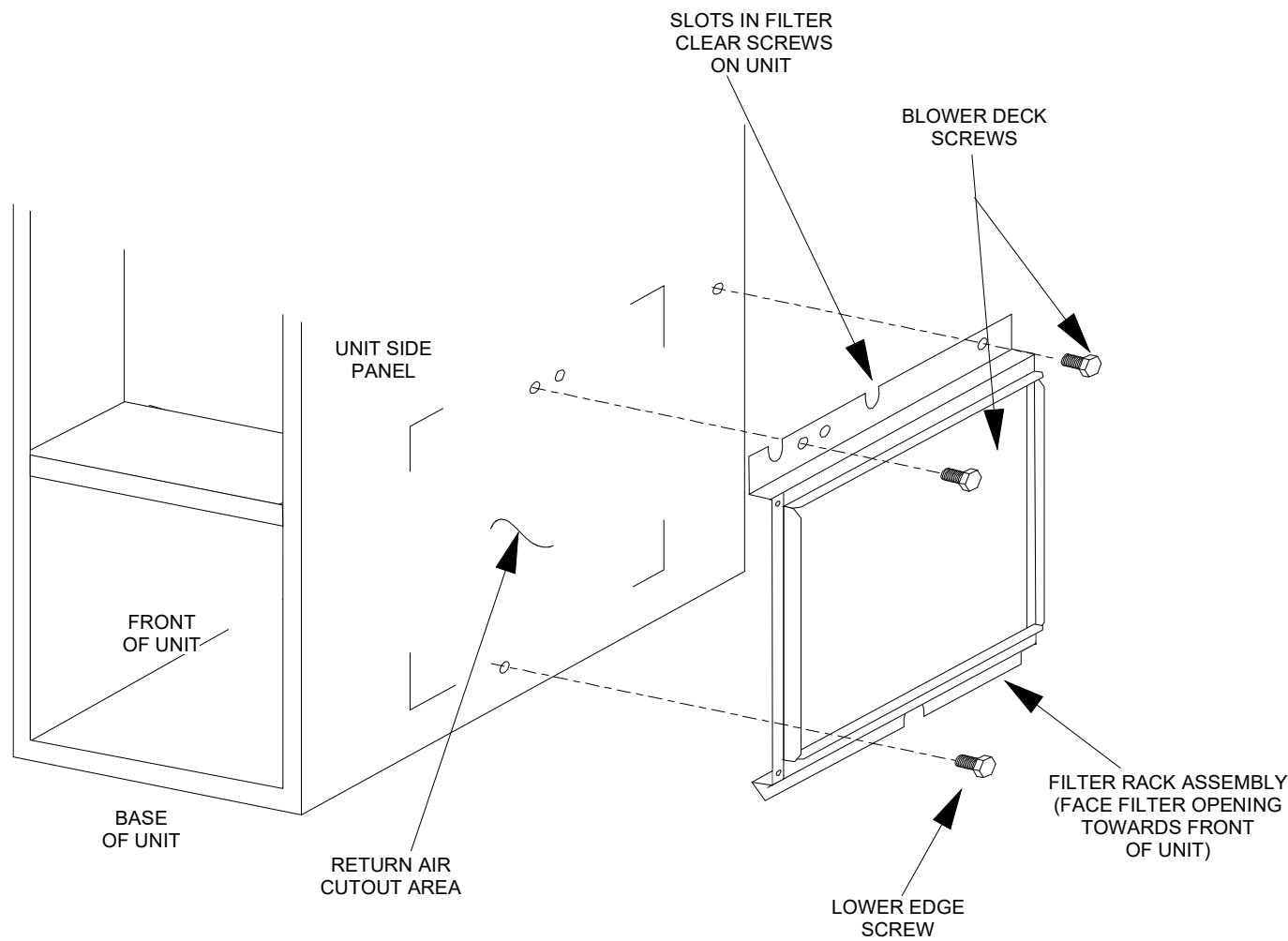
1. R and G thermostat contacts close, initiating a call for fan.
2. Integrated control module performs safety circuit checks.
3. Circulator blower is energized on low heat speed. Electronic air cleaner terminals are energized.
4. Circulator blower runs, integrated control module monitors safety circuits continuously.
5. R and G thermostat contacts open, completing the call for fan.
6. Furnace awaits the next call from thermostat.

Daikin "S" Model Furnace Accessories																
MODEL NUMBER	EFR02	AFE18-60A	AMU / GMU	ASAS / GSAS	0170K00000S	0170K00001S	DCVK-20 (CVENT-2)	DCVK-30 (CVENT-3)	LPM08	LPLP03	RF000142	TK-400	HASFK-04	HASFK-04	HASFK-05	HASFK-06
Description	EFR External Filter Rack	Dual Fuel Kit	Media Air Cleaners	Electronic Air Cleaner	Concentric Side Wall Flush Mount Vent Kit (3")	Concentric Side Wall Flush Mount Vent Kit (2")	Concentric Vent Kit (2")	Concentric Vent Kit (3")	Propane Gas Conversion Kit	LP Low Pressure Shut Off Kit	Vent Drain Coupling	Twinning Kit	High Altitude Natural Gas Kit	High Altitude LP Gas Kit	High Altitude LP Gas Kit	High Altitude LP Gas Kit
DM96SE0403ANA*	•	•	•	•	•	•	•	•	•	•	•	•	1	1		
DM96SE0603BNA*	•	•	•	•	•	•	•	•	•	•	•	•	1			1
DM96SE0803BNA*	•	•	•	•	•	•	•	•	•	•	•	•	1			1
DM96SE0804CNA*													1			1
DM96SE0805CNA*													1		1	
DM96SE1005CNA*	•	•	•	•	•	•	•	•	•	•	•	•	1		1	
DM96SE1205DNA*	•	•	•	•	•	•	•	•	•	•	•	•	1		1	
DC96SE0403BNA*		•	•	•	•	•	•	•	•	•	•	•	1		1	
DC96SE0603BNA*		•	•	•	•	•	•	•	•	•	•	•	1		1	
DC96SE0804CNA*		•	•	•	•	•	•	•	•	•	•	•	1		1	
DC96SE1005CNA*		•	•	•	•		•	•	•	•	•	•	1	1		
DC96SE1205DNA*		•	•	•	•		•	•	•	•	•	•	1	1		

1 = above 7,000 altitude

ACCESSORIES

EXTERNAL FILTER RACK KIT (EFR02)



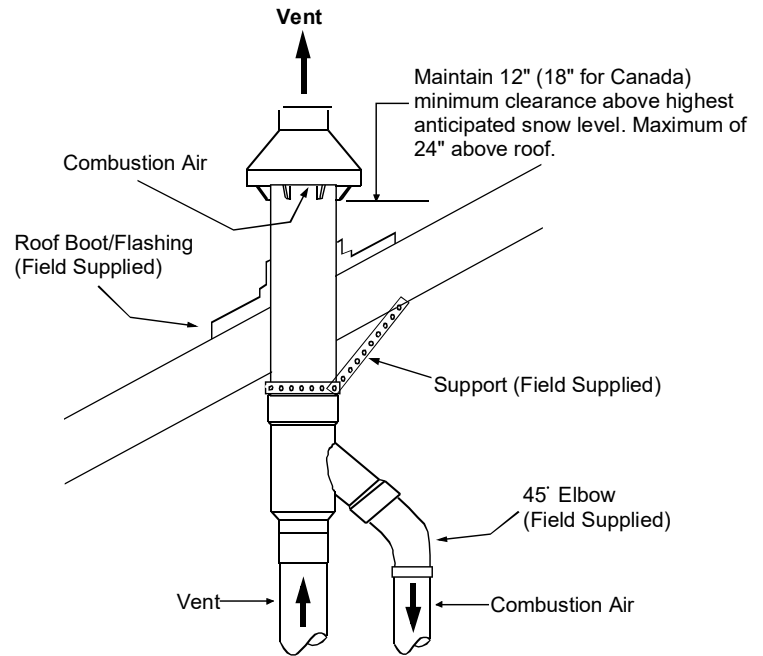
EFR02 EXTERNAL FILTER RACK KIT

90% Upflow Model Furnaces

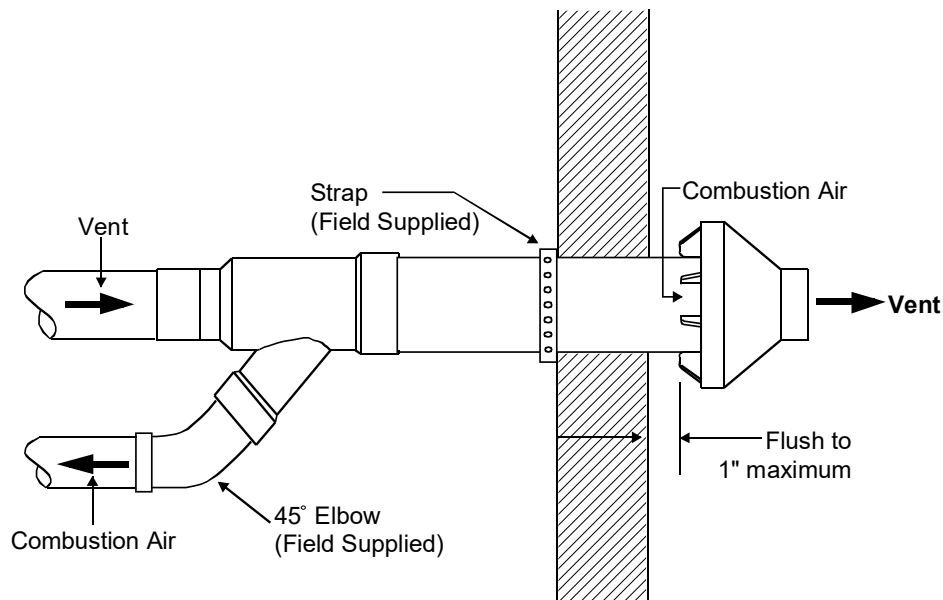
CONCENTRIC VENT CONVERSION KIT (DCVK-20 / CVENT-2) (DCVK-30 / CVENT-3)

The DCVK-20 (2") or the DCVK-30 (3") is a concentric vent kit approved with furnaces listed in this manual. This concentric vent kit allows for vertical or horizontal vent termination. The illustrations give a brief view of the kit and its application.

See the section in this manual under "Vent Flue and Combustion Air Pipe Terminations" for more information or consult the Installation and Operating Instructions (IO-619*).



DCVK (VERTICAL INSTALLATION)



DCVK (HORIZONTAL INSTALLATION)

ACCESSORIES

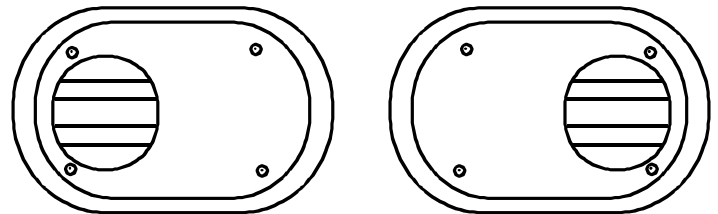
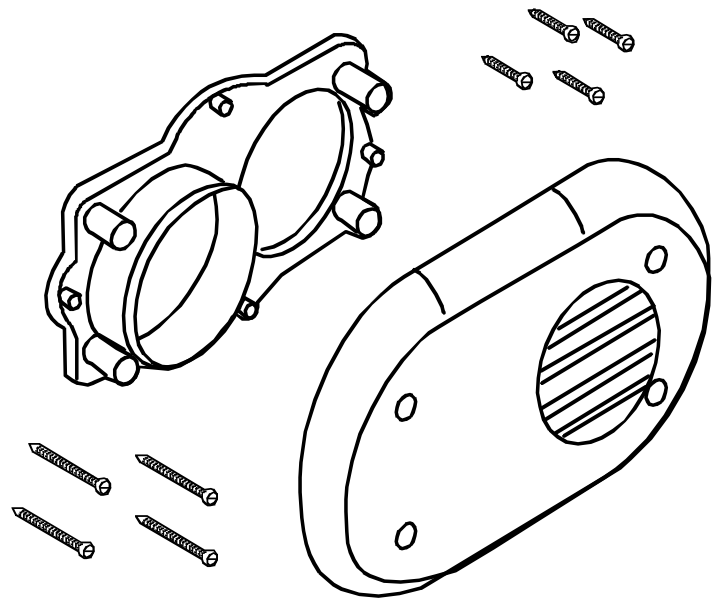
SIDE WALL VENT KIT (0170K00000S / 0170K00001S)

DESCRIPTION

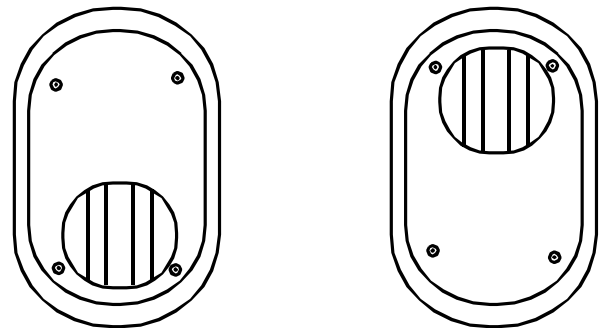
These side wall only vent kits are to be used with 2" - 3" vent systems. These kits are NOT intended for use with single pipe (indirect vent) installations.

These vent kits must terminate outside the structure and may be installed with the intake and exhaust pipes located side-by-side or with one pipe above the other.

See the section in this manual under "Vent Flue and Combustion Air Pipe Terminations" for more information or consult the Installation Instructions (IO-635* or IO-805).



Horizontal Installation



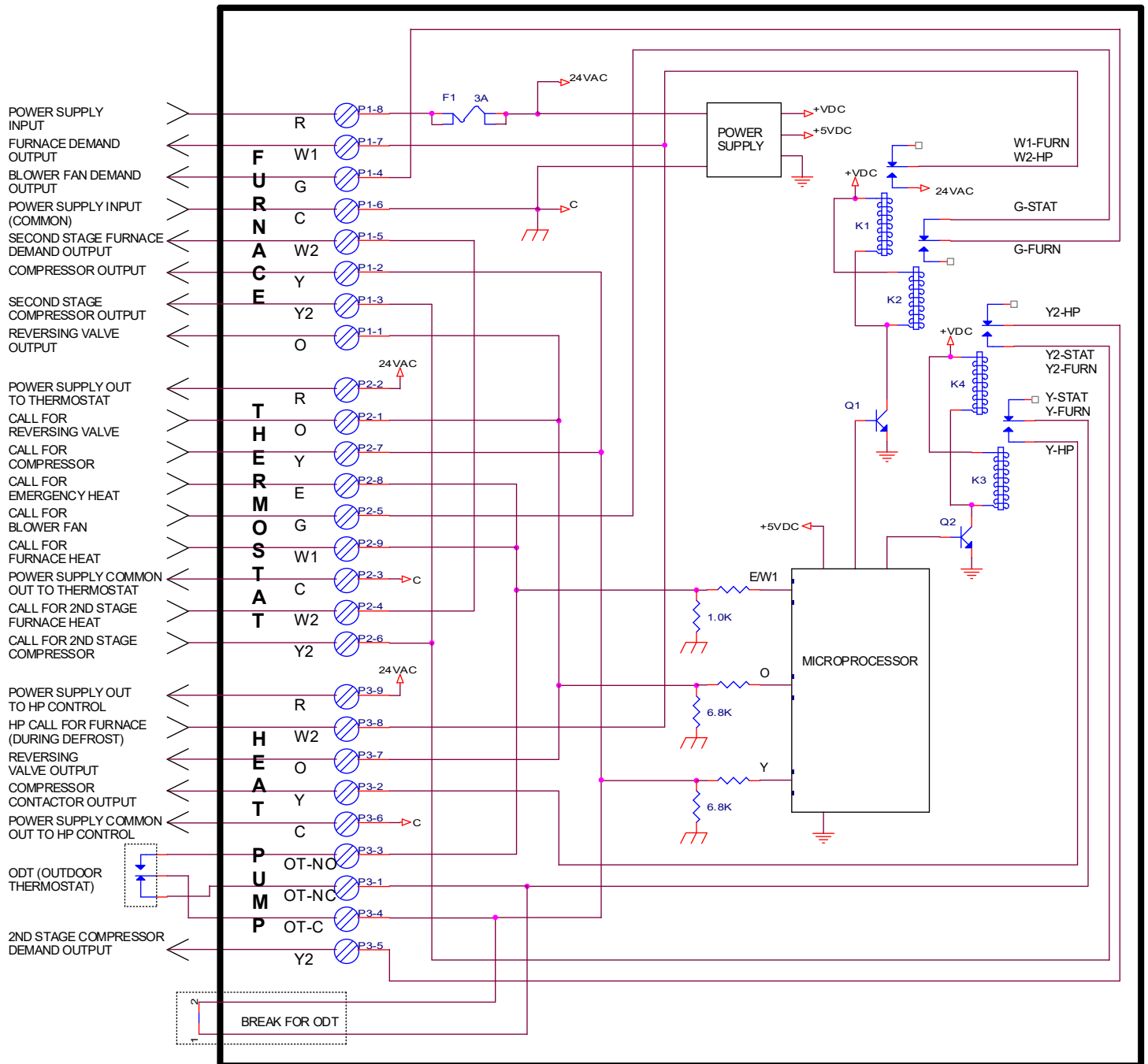
Vertical Installation



WARNING

TO AVOID POSSIBLE ELECTRICAL SHOCK, PERSONAL INJURY, OR DEATH, DISCONNECT THE POWER BEFORE SERVICING.

ALL FUEL SYSTEM CONTROL BOARD - AFE1860A



ALL FUEL SYSTEM CONTROL BOARD - AFE18-60A

This wiring diagram is for reference only. Not all wiring is as shown above.

Refer to the appropriate wiring diagram for the unit being serviced.

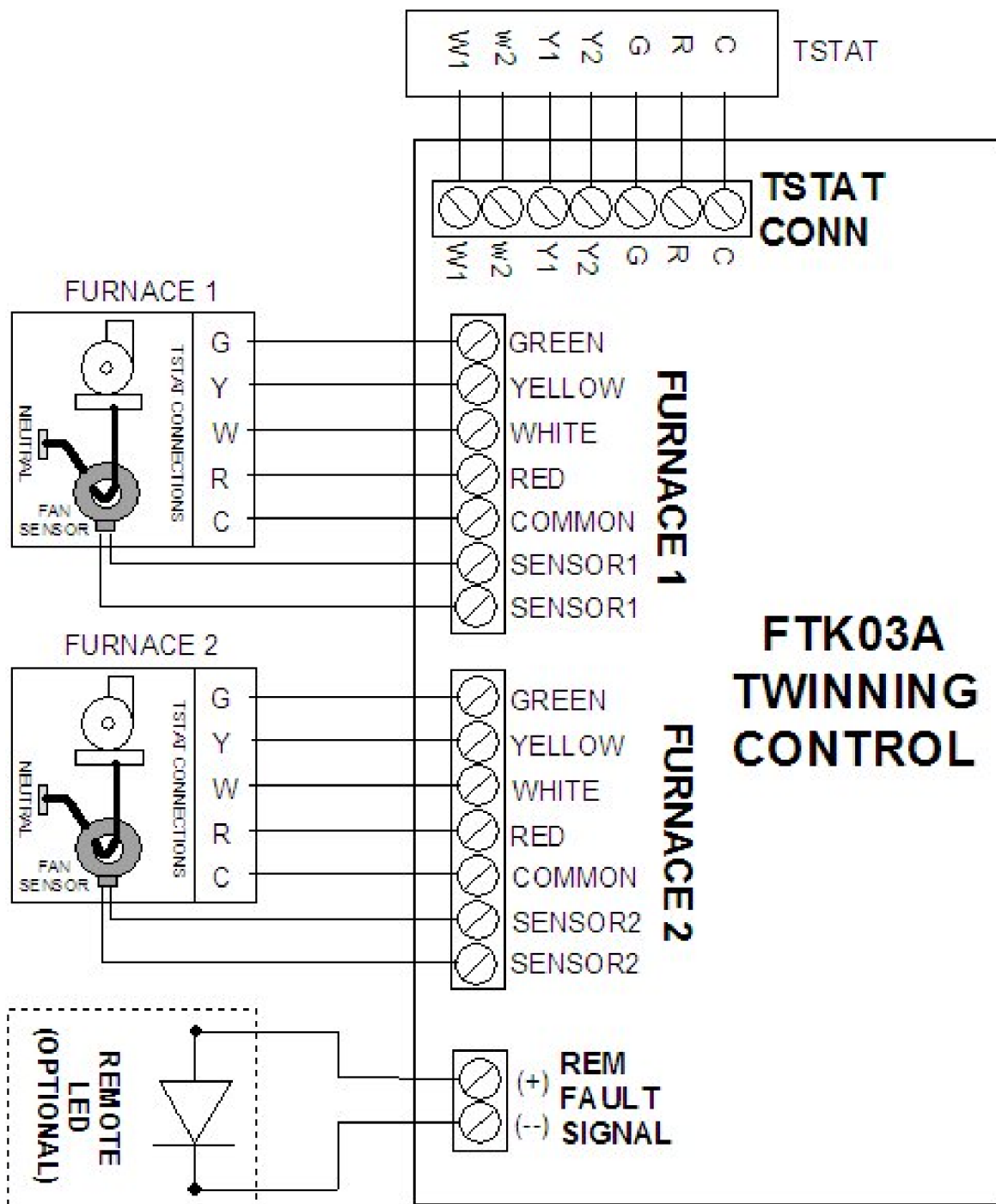
(For use with Heat Pumps in conjunction with 80% or 90% Single-Stage or Two-Stage Furnaces)

ACCESSORY WIRING DIAGRAMS



WARNING

TO AVOID POSSIBLE ELECTRICAL SHOCK, PERSONAL INJURY, OR DEATH, DISCONNECT THE POWER BEFORE SERVICING.



FURNACE TWINNING KIT - FTK04 (UTEC Control Board)

This wiring diagram is for reference only. Not all wiring is as shown above.

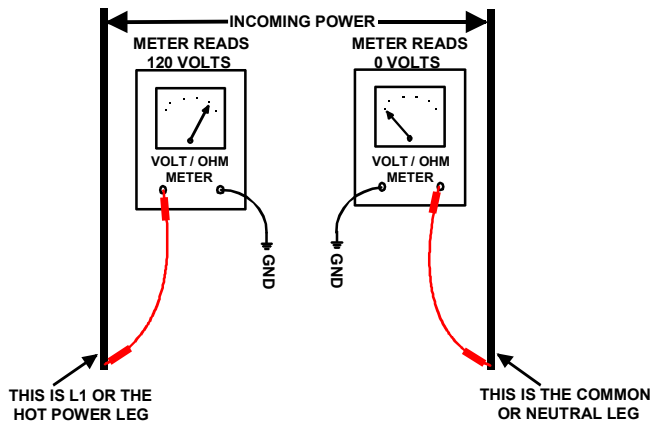
Refer to the appropriate wiring diagram for the unit being serviced.

(For use only with 80% or 90% Single-Stage Furnaces)

As more and more electronic's are introduced to the Heating Trade, Polarization of incoming power and phasing of primary to secondary voltage on transformers becomes more important.

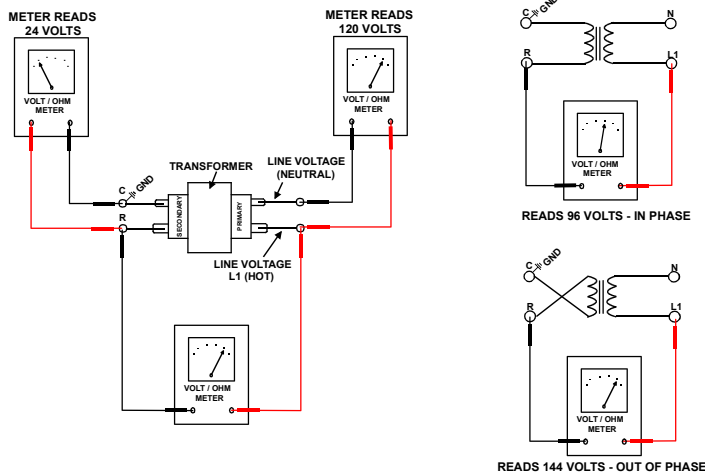
Polarization has been apparent in the Appliance industry since the introduction of the three prong plug, however, the Heating Industry does not use a plug for incoming power, but is hard wired.

Some of the electronic boards being used today, with flame rectification, will not function properly and/or at all without polarization of incoming power. Some also require phasing between the primary and secondary sides of step-down transformers.



These then should be wired to the furnace accordingly.

CHECKING FOR PHASING - PRIMARY TO SECONDARY OF UNMARKED TRANSFORMERS*



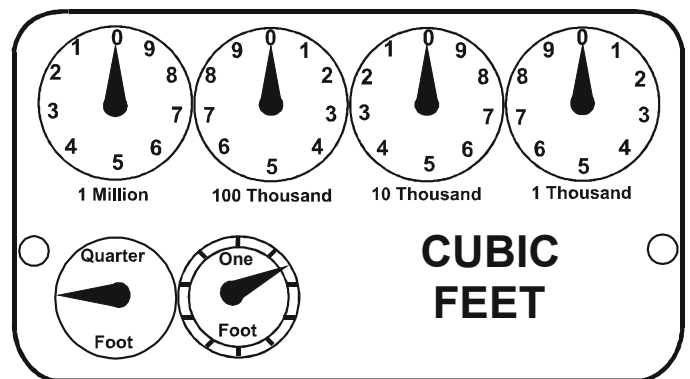
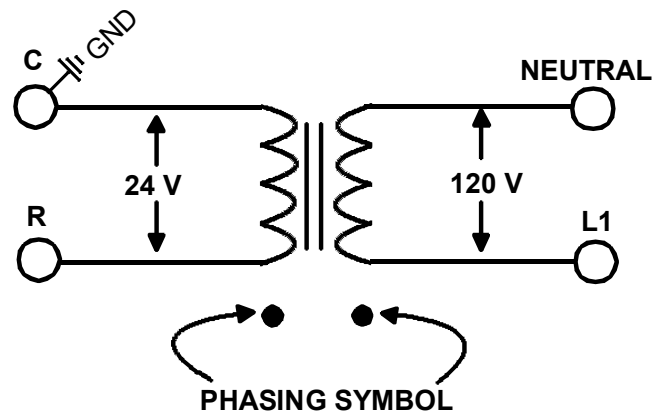
If meter reads approximately 96 volts - the primary to secondary are in phase - if reads approximately 144 volts out of phase - reverse low voltage wires.

***NOTE: FOR FLAME RECTIFICATION THE COMMON SIDE OF THE SECONDARY VOLTAGE (24 V) IS CABINET GROUNDED. IF YOU WERE TO BENCH TEST A TRANSFORMER THE PRIMARY NEUTRAL AND SECONDARY COMMON MUST BE CONNECTED TOGETHER FOR TESTING PURPOSES.**

Some transformers will display phasing symbols as shown in the illustration to the left to assist in determining proper transformer phasing.

Checking for polarization and phasing should become a habit in servicing. Let's start now.

NOTE: NEWER INTEGRATED IGNITION CONTROLS HAVE A DIAGNOSTIC FLASH CODE FOR REVERSED POLARITY (REFER TO TROUBLESHOOTING-DIAGNOSTIC CHART FOR LED CODES).



SERVICING

GAS RATE -- CUBIC FEET PER HOUR					
Seconds for One Revolution	Size of Test Dial				
	1/4 cu/ft	1/2 cu/ft	1 cu/ft	2 cu/ft	5 cu/ft
10	90	180	360	720	1800
11	82	164	327	655	1636
12	75	150	300	600	1500
13	69	138	277	555	1385
14	64	129	257	514	1286
15	60	120	240	480	1200
16	56	113	225	450	1125
17	53	106	212	424	1059
18	50	100	200	400	1000
19	47	95	189	379	947
20	45	90	180	360	900
21	43	86	171	343	857
22	41	82	164	327	818
23	39	78	157	313	783
24	37	75	150	300	750
25	36	72	144	288	720
26	34	69	138	277	692
27	33	67	133	265	667
28	32	64	129	257	643
29	31	62	124	248	621
30	30	60	120	240	600
31	--	--	116	232	581
32	28	56	113	225	563
33	--	--	109	218	545
34	26	53	106	212	529
35	--	--	103	206	514
GAS RATE -- CUBIC FEET PER HOUR					
Seconds for One Revolution	Size of Test Dial				
	1/4 cu/ft	1/2 cu/ft	1 cu/ft	2 cu/ft	5 cu/ft
36	25	50	100	200	500
37	--	--	97	195	486
38	23	47	95	189	474
39	--	--	92	185	462
40	22	45	90	180	450
41	--	--	--	176	439
42	21	43	86	172	429
43	--	--	--	167	419
44	--	41	82	164	409
45	20	40	80	160	400
46	--	--	78	157	391
47	19	38	76	153	383
48	--	--	75	150	375
49	--	--	--	147	367
50	18	36	72	144	360
51	--	--	--	141	355
52	--	--	69	138	346
53	17	34	--	136	340
54	--	--	67	133	333
55	--	--	--	131	327
56	16	32	64	129	321
57	--	--	--	126	316
58	--	31	62	124	310
59	--	--	--	122	305
60	15	30	60	120	300

COMBUSTION QUALITY

Combustion quality can be affected by several factors. Major factors are venting and draining.

VENTING

The venting system could be planned and installed with the following in mind:

- Should not be longer than necessary
- Use 45° elbows rather than 90° elbows when possible
- Must not sag or otherwise trap condensate
- Use longest radius fittings possible
- If using 3" venting, make the transition from 2" to 3" as close as practically possible
- Make sure there is no flue gas recirculation into the combustion air pipe

CONDENSATE DRAINAGE

Furnace combustion can be affected if a furnace is holding condensate. Check for proper connections of drain hoses; make sure furnace condensate trap is clean. Make sure furnace is not improperly sloped. Make sure air conditioning coil drain is not interfering with furnace drain.

OTHER CAUSES

1. Manifold Gas Pressure must be set, the gas being used, Natural or L.P., high and low firing rates. If converted to L.P. gas, check size of all orifices.
2. Remove Draft Inducer, Check the integrity of the gasket between the inducer and the collector box cover, any air leak here will have a negative effect on combustion. Check the orifice hole in the collector box, it must be free or burrs on both sides.
3. Make sure burners are clean, not out of position and line up correctly with heat exchanger tubes, including the heat exchanger orifice plate - this is the plate which mounts to the front panel of the heat exchanger between the burners and the heat exchanger tubes, make sure it is not loose, missing a screw or hanging down between the burners and heat exchanger causing flame impingement.
4. Make sure the field installed gas line is not binding and causing distortion of burner assembly.
5. If the furnace is installed as a one pipe system; make sure the surrounding area and structure are adequate to provide combustion air.
6. Make sure there are no cabinet air leaks allowing supply air to affect combustion.
7. If heat exchanger integrity is uncertain, follow procedures in Service Bulletin SF-041.

Service Problem		No Heat			Unsatisfactory Heat				Test Method Remedy	See Service Procedure Reference			
POSSIBLE CAUSE DOTS IN ANALYSIS GUIDE INDICATE "POSSIBLE CAUSE"	SYMPTOM	System Will Not Start	Burner Won't Ignite	Burner Ignites-Locks Out	Burner Shuts Off prior to T'Stat being Satisfied	Short Cycles	Long Cycles	Soot and /or Fumes			Too Much Heat	Not Enough Heat	
Power Failure	•										Test Voltage	S-1	
Blown Fuse	•										Test Voltage	S-4	
Loose Connection	•										Check Wiring	S-2	
Shorted or Broken Wires	•										Check Wiring	S-3A	
No Low Voltage	•										Check Transformer	S-4	
Faulty Thermostat	•					•	•		•	•	Check Thermostat	S-3A	
Faulty Transformer	•										Check Transformer	S-4	
Poor or High Resistance Ground				•							Measure Ground Resistance	S-17B	
Improper Heat Anticipator Setting						•	•		•	•	Adjust Heat Anticipator Setting	S-3B	
Improper Thermostat Location						•	•		•	•	Relocate Thermostat	S-316	
Faulty Limit or Roll Out Switch			•		•						Test Control	S-300 / S-302	
Faulty Flame Sensor				•							Test Flame Sensor	S-314	
Faulty Ignition Control	•	•	•	•							Test Control	S-313	
Gas Valve or Gas Supply Shut Off			•								Turn Valves to On Position	S-317	
Faulty Induced Draft Blower			•						•		Test Induced Draft Motor	S-303	
Faulty Blower Motor (PSC)			•		•						Test Blower Motor (PSC)	S-16A,B,C	
Broken or Shorted Ignitor			•								Test Ignitor	S-308	
Dirty Flame Sensor, Low UA				•							Clean Flame Sensor	S-314	
Stuck Gas Valve			•						•	•	Replace Gas Valve	S-304	
Faulty Gas Valve			•						•	•	Replace Gas Valve	S-304	
Open Auxiliary Limit			•		•						Reset Control	S-301	
Improper Air Flow or Distribution					•					•	Check Duct Static	S-200	
Cycling on Limit					•	•				•	Check Controls & Temperature Rise	S-201 / S-300	
Delayed Ignition									•		Test for Delayed Ignition	S-312	
Flashback									•		Test for Flashback	S-309 / S-311	
Orifice Size									•	•	•	Check Orifices	S-306 / S-311
Gas Pressure			•						•	•	•	Check Gas Pressure	S-307
Cracked Heat Exchanger									•		Check Burner Flames	S-305	
Furnace Undersized							•			•	Replace with Proper Size Furnace	S-318	
Furnace Oversized						•					Replace with Proper Size Furnace	S-318	
Faulty Pressure Switch			•	•	•						Test Pressure Switch	S-310	
Blocked or Restricted Flue									•		Check Flue/Draw down Pressure	S-310	
Open Roll Out Switch			•	•							Test Flame Roll Out Control	S-302	
Bouncing On Pressure Switch				•							Test Negative Pressure	S-310	

TROUBLESHOOTING

Troubleshooting Chart DM96SE, DC96SE

Symptoms of Abnormal Operation	Associated LED Code ²	Fault Description(s)	Possible Causes	Corrective Action	Cautions and Notes
- Furnace fails to operate. - Integrated control module diagnostic LED provides no signal.	OFF	- No 115 volt power to furnace, or no 24 volt power to integrated control module. - Blown fuse or circuit breaker. - Integrated control module has an internal fault.	- Manual disconnect switch OFF, door switch open, or 24 volt wires improperly connected or loose. - Blown fuse or circuit breaker. - Integrated control module has an internal fault.	- Assure 115 and 24 volt power to furnace integrated control module. - Check integrated control module fuse (3A). Replace if necessary. - Check for possible shorts in 115 and 24 volt circuits. Repair as necessary. - Replace bad integrated control module.	- Turn power OFF prior to repair. - Replace integrated control module fuse with 3A automotive fuse. - Read precautions in "Electrostatic Discharge" section of manual.
LED is Steady On.	 CONTINUOUS ON	Normal Operation	Normal Operation	None	Normal Operation
- Furnace fails to operate. - Integrated control module diagnostic LED is flashing ONE (1) flash.	 1 FLASH	Furnace lockout due to an excessive number of ignition "retries" (3 total)¹.	- Failure to establish flame. Cause may be no gas to burners, front cover pressure switch stuck open, bad igniter or igniter alignment, improper orifices, coated/oxidized or improperly connected flame sensor. - Loss of flame after establishment. Cause may be interrupted gas supply, lazy burner flames (improper gas pressure or restriction in flue and/or combustion air piping), front cover pressure switch opening, improper induced draft blower performance.	- Locate and correct gas interruption. - Check front cover pressure switch operation (hose, wiring, contact operation). Correct if necessary. - Replace or realign igniter. - Check flame sense signal. Sand sensor if coated and/or oxidized. - Check flue piping for blockage, proper length, elbows, and termination. - Verify proper induced draft blower performance.	- Turn power OFF prior to repair. - Igniter is fragile, handle with care. - Sand flame sensor with emery cloth. - See "Vent/ Flue Pipe" section for piping details.
- Furnace fails to operate. - Integrated control module diagnostic LED is flashing TWO (2) flashes.	 2 FLASHES	- Pressure switch circuit is closed. - Induced draft blower is not operating.	- Induced draft blower pressure switch contacts sticking. - Shorts in pressure switch circuit.	- Replace induced draft blower pressure switch. - Repair short.	- Turn power OFF prior to repair. - Replace pressure switch with proper replacement part.
- Induced draft blower runs continuously with no further furnace operation. - Integrated control module diagnostic LED is flashing THREE (3) flashes.	 3 FLASHES	- Pressure switch circuit not closed. - Induced draft blower is operating.	- Pressure switch hose blocked, pinched or connected improperly. - Blocked flue and/or inlet air pipe, blocked drain system, or weak induced draft blower. - Incorrect pressure switch setpoint or malfunctioning switch contacts. - Loose or improperly connected wiring.	- Inspect pressure switch hose. Repair, if necessary. - Inspect flue and/or inlet air piping for blockage, proper length, elbows, and termination. Check drain system. Correct as necessary. - Correct pressure switch setpoint or contact motion. - Tighten or correct wiring connection.	- Turn power OFF prior to repair. - Replace pressure switch with proper replacement part.

TROUBLESHOOTING

• Circulator blower runs continuously. No furnace operation. • Integrated control module diagnostic LED is flashing FOUR (4) flashes.	 4 FLASHES	• Primary or auxiliary limit circuit is open.	• Faulty primary or auxiliary limit switch. • Insufficient conditioned air over the heat exchanger. Blocked filters, restrictive ductwork, improper circulator blower speed, or failed circulator blower. • Loose or improperly connected wiring.	• Check primary/auxiliary limit. Replace if necessary. • Check filters and ductwork for blockage. Clean filters or remove obstruction. • Check circulator blower speed and performance. Correct speed or replace blower if necessary. • Tighten or correct	• Turn power OFF prior to repair. • Replace primary/auxiliary limit with proper replacement part. • Replace blower with correct replacement part.
• Induced draft blower and circulation blower runs continuously. No furnace operation. • Integrated control module diagnostic LED is flashing FIVE (5) flashes.	 5 FLASHES	• Flame sensed with no call for heat.	• Short to ground in flame sense circuit.	• Correct short at flame sensor or in flame sensor wiring.	• Turn power OFF prior to repair.
• Furnace fails to operate. • Integrated control module diagnostic LED is flashing SIX (6) flashes. • No furnace operation.	 6 FLASHES	• Rollout limit open. • Integrated control module fuse is blown.	• Flame rollout. • Misaligned burners, blocked flue and/or air inlet pipe, or failed induced draft blower. • Loose or improperly connected wiring. • Short in 24 volt AC control circuits or safety circuits. • Faulty rollout limit.	• Check burners for proper alignment. • Check flue and air inlet piping for blockage, proper length, elbows, and termination. Correct as necessary. • Check rollout limit. Replace if necessary. • Check induced draft blower for proper performance. Replace, if necessary. • Tighten or correct wiring connection.	• Turn power OFF prior to repair. • See "Vent/Flue Pipe" section for piping details. • Replace induced draft blower with correct replacement part. • Replace integrated control module fuse with 3A automotive fuse. • Read precautions in "Electrostatic Discharge" section of manual. • Replace rollout limit with correct replacement part.
• Normal furnace operation. • Integrated control module diagnostic LED is flashing SEVEN (7) flashes.	 7 FLASHES	• Flame sense microamp signal is low.	• Flame sensor is coated/oxidized. • Flame sensor incorrectly positioned in burner flame. • Lazy burner flame due to improper gas pressure or combustion air.	• Sand flame sensor. • Inspect for proper sensor alignment. • Compare current gas pressure to rating plate info. Adjust as needed.	• Turn power OFF prior to repair. • Clean flame sensor with steel wool. • See "Vent/Flue Pipe" section for piping details. • See rating plate for proper gas pressure.
• Furnace not operating. • Integrated control module diagnostic LED is flashing EIGHT (8) flashes.	 8 FLASHES	• Igniter Relay Fault	• Improperly connected igniter. • Bad igniter. • Poor unit ground. • Poor burner ground. • Faulty integrated control module.	• Check and correct wiring from integrated control module to igniter. • Replace bad igniter. • Check & correct unit ground wire. • Replace bad integrated control module.	• Turn power OFF prior to repair. • Replace igniter w/proper replacement part. • Read precautions in "Electrostatic Discharge" section of manual.
• Circulating blowers not operating simultaneously. No blower operation.	 9 FLASHES	• Twinning Fault	• Furnace twin terminals not connected together by field wiring. • Isolating relay not used or improperly installed	• Install field wiring connecting twin terminals, check continuity of wiring. • Verify each furnace fan motor is operable without twinning connection. • Wire per twinning diagram in this installation manual	• Verify line voltage supply is turned off before proceeding with corrections

TROUBLESHOOTING

- Furnace not operating. - Integrated control module diagnostic LED is flashing TEN (10) flashes.	 10 FLASHES	Open Fuse / No 24V power to integrated control module	High voltage surge blows off the fuse.	Replace the Fuse.	Verify line voltage supply is turned off before proceeding with corrections
- Furnace not operating. - Integrated control module diagnostic LED is flashing ELEVEN (11) flashes.	 11 FLASHES	Igniter Open	- Improperly connected igniter. - Bad igniter. - Poor unit ground. - Poor burner ground.	- Check and correct wiring from integrated control module to igniter. - Replace bad igniter. - Check & correct unit ground wire. - Replace bad integrated	- Turn power OFF prior to repair. - Replace igniter w/proper replacement part. - Read precautions in "Electrostatic Discharge" section
- Furnace not operating. - Integrated control module diagnostic LED is flashing TWELVE (12) flashes.	 12 FLASHES	Inducer Relay Fault	- Improperly connected inducer. - Bad inducer. - Poor unit ground. - Faulty integrated control module.	- Check and correct wiring from integrated control module to inducer. - Check for Blockage in inducer & in Vent / replace Inducer. - Check & correct unit ground wire. - Replace bad integrated control module.	- Turn power OFF prior to repair. - Replace inducer w/proper replacement part. - Read precautions in "Electrostatic Discharge" section of manual.
- Furnace fails to operate. - Integrated control module diagnostic LED is flashing continuously.	 CONTINUOUS RAPID FLASH	Polarity of 115 or 24 volt power is reversed.	- Polarity of 115 volt AC power to furnace or integrated control module is reversed. - Orange and gray wires to transformer are reversed. - Poor unit ground.	- Review wiring diagram to correct polarity. - Reverse orange and gray wires going to transformer. - Verify proper ground. Correct if necessary.	Turn power OFF prior to repair

¹Integrated control module will automatically attempt to reset from lockout after one hour.

²LED Flash code will cease if power to the control module is interrupted through the disconnect or door switch.

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4. With ample voltage at line voltage connectors, energize the furnace blower motor by jumpering terminals R to G on the integrated ignition control.
5. With the blower motor in operation, the voltage should be 115 volts $\pm$ 10 percent.
6. If the reading falls below the minimum voltage, check the line wire size. Long runs of undersized wire can cause low voltage. If wire size is adequate, notify the local power company of the condition.
7. After completing check and/or repair, replace Junction Box cover and reinstall the service panel doors.
8. Turn on electrical power and verify proper unit operation.

S-2 CHECKING WIRING



WARNING

DISCONNECT ALL POWER BEFORE SERVICING.

1. Check wiring visually for signs of overheating, damaged insulation and loose connections.
2. Use an ohmmeter to check continuity of any suspected open wires.
3. If any wires must be replaced, replace with AWM, 105°C. 2/64 thick insulation of the same gauge or its equivalent.

The heating anticipator setting for furnaces covered in this manual is **0.70 Amps**.

COOLING ANTICIPATOR

The cooling anticipator is a small heater (resistor) in the thermostat. During the "OFF" cycle it heats the bimetal element helping the thermostat call for the next cooling cycle. This prevents the room temperature from rising too high before the system is restarted. A properly sized anticipator should maintain room temperature within 1 1/2 to 2 degrees.

The anticipator is fixed in the subbase and is not to be replaced. If the anticipator should fail for any reason, the subbase must be changed.

S-4 CHECKING TRANSFORMER AND CONTROL CIRCUIT

A step-down transformer 120 volt primary to 24 volt secondary, 40 VA (Heating and Cooling Models) supplies ample capacity of power for either operation.

S-1 CHECKING VOLTAGE



WARNING

HIGH VOLTAGE

DISCONNECT ALL POWER BEFORE SERVICING OR CHANGING ANY ELECTRICAL WIRING. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.



1. Remove cover from the Junction Box and gain access to incoming power lines.

With Power ON:



WARNING

LINE VOLTAGE NOW PRESENT

2. Using a voltmeter, measure the voltage across the hot and neutral connections.

NOTE: To ENERGIZE THE FURNACE, THE DOOR INTERLOCK SWITCH MUST BE ENGAGED AT THIS POINT.

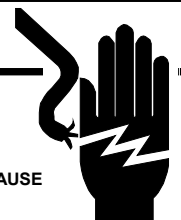
3. No reading - indicates open wiring, open fuse, no power, or faulty Door Interlock Switch from unit to fused disconnect service. Repair as needed.



WARNING

HIGH VOLTAGE

DISCONNECT ALL POWER BEFORE SERVICING OR CHANGING ANY ELECTRICAL WIRING. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.



SERVICING

1. Remove blower compartment door to gain access to the thermostat low voltage wires located at the furnace integrated control module.
2. Remove the thermostat low voltage wires at the furnace integrated control module terminals.

With Power On (and Door Interlock Switch closed):



WARNING

LINE VOLTAGE NOW PRESENT

3. Use a voltmeter, check voltage across terminals R and C. Must read 24 VAC.
4. No voltage indicates faulty transformer, open fuse, bad wiring, bad splice, or open door interlock switch.
5. Check transformer primary voltage at incoming line voltage connections, fuse, splices, and blower door interlock switch.
6. If line voltage is available to the primary side of transformer and not at secondary side, the transformer is inoperative. Replace.
7. After completing check and/or replacement of transformer and check and/or repair of control circuit, reinstall blower compartment door.
8. Turn on electrical power and verify proper unit operation.

S-10B CAPACITANCE CHECK



WARNING

LINE VOLTAGE NOW PRESENT



WARNING

HIGH VOLTAGE

DISCONNECT **ALL** POWER BEFORE SERVICING OR INSTALLING THIS UNIT. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.



1. Remove blower compartment door to gain access to capacitor.
2. Discharge capacitor and remove wire leads.
3. Once the capacitor is discharged, use a capacitor tester to determine the condition of the capacitor. A good capacitor will have a reading within 5% of the rated value.

S-16A CHECKING AIR CIRCULATOR BLOWER MOTOR



WARNING

DISCONNECT **ALL** POWER BEFORE SERVICING.

1. Remove blower compartment door to gain access to the circulator blower motor and integrated ignition control.
2. Check for any obstruction that would keep the fan wheel / fan motor from turning.
3. Check wiring, the motor has two wiring harnesses, a main harness and a control harness. The main pin harness has:

White neutral wire connected to the Neutral terminal on the control board.

Black wire connected to the CIRC H terminal on the control board.

Red wire connected to the COM terminal, which is a female spade connection next to the T1 – T4 wires on the control board.

Green ground wire connected to cabinet ground

The control harness has:

Blue wire connected to T1 on the control board.

Red wire connected to T2 on the control board.

Orange wire connected to T3 on the control board.

Black wire connected to T4 on the control board.

The multi-speed ECM motor requires a line voltage power supply (black connected to CIRC H and white connected to neutral on the control board) as well as a signal on one of the speed taps (T1-T4).

The speed tap voltage is A.C. and can vary which tap is energized depending on DIP switch selection. The voltage reading from any one of the speed taps is referenced between the female COM terminal next to the speed taps on the control board. From COM to T1 or T2, T3, T4, you should read 24 VAC on the low voltage speed taps depending on Dip switch settings.

SERVICING

Motor Tap Identification		
CONNECTOR ID	DESCRIPTION	CONNECTOR VOLTAGE
L	LINE, L1	LINE, L1
G	GROUND	CHASSIS GROUND
N	LINE, L2	LINE, L2
C	SIGNAL COMMON	24VAC COMMON
1	TAP 1	24VAC
2	TAP 2	24VAC
3	TAP 3	24VAC
4	TAP 4	24VAC
5	TAP 5	24VAC

Heat Off Delay (S1-4)				
S1-4	Min	Nom	Max	Units
ON		*150		Seconds
OFF		100		Seconds
* = DEFAULT SETTING				

COOL MOTOR SPEED DIP SWITCH SETTING (S1-1, S1-2, S1-3)										
	Position	1 2 3	1 2 3	1 2 3	1 2 3	*1 2 3	1 2 3	1 2 3	1 2 3	1 2 3
Motor Speed COOL	Y	T1	T2	T3	T4	T1+T2	T1	T1	T1	T1
* = DEFAULT SETTING										

HEAT MOTOR SPEED DIP SWITCH SETTING (S2-1, S2-2)					
		1 2	*1 2	1 2	1 2
	Position	OFF OFF	ON OFF	ON ON	OFF ON
Motor Speed HEAT	W	T1	T2	T3	T4
* = DEFAULT SETTING					

FAN MOTOR SPEED DIP SWITCH SETTING (S2-3, S2-4)					
		*3 4	3 4	3 4	3 4
	Position	OFF OFF	ON OFF	OFF ON	ON ON
Motor Speed FAN	G	T1	T2	T4	T1+T2
* = DEFAULT SETTING					

S-200 CHECKING DUCT STATIC

The maximum and minimum allowable external static pressures are found in the specification section. These tables also show the amount of air being delivered at a given static by a given motor speed or pulley adjustment. The furnace motor cannot deliver proper air quantities (CFM) against statics other than those listed.

Too great of an external static pressure will result in insufficient air that can cause excessive temperature rise, resulting in limit tripping, etc. Whereas not enough static may result in motor overloading.

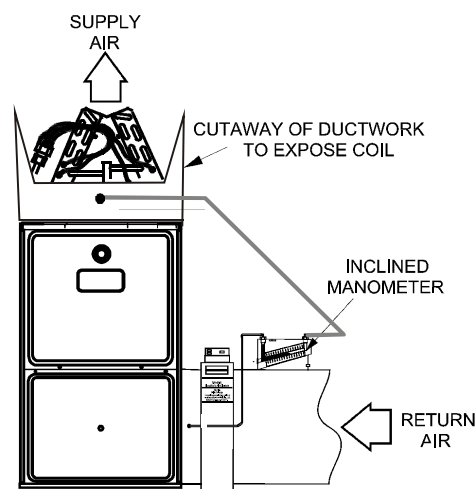
To determine proper air movement, proceed as follows:

1. With clean filters in the furnace, use a draft gauge (inclined manometer) to measure the static pressure of the return duct at the inlet of the furnace. (Negative Pressure)
2. Measure the static pressure of the supply duct. (Positive Pressure)
3. Add the two (2) readings together for total external static pressure.

NOTE: BOTH READINGS MAY BE TAKEN SIMULTANEOUSLY AND READ DIRECTLY ON THE MANOMETER IF SO DESIRED. IF AN AIR CONDITIONER COIL OR ELECTRONIC AIR CLEANER IS USED IN CONJUNCTION WITH THE FURNACE, THE READINGS MUST ALSO INCLUDE THESE COMPONENTS, AS SHOWN IN THE FOLLOWING DRAWING.

4. Consult proper tables for the quantity of air.

If the total external static pressure exceeds the minimum or maximum allowable statics, check for closed dampers, registers, undersized and/or oversized poorly laid out duct work.



CHECKING STATIC PRESSURE

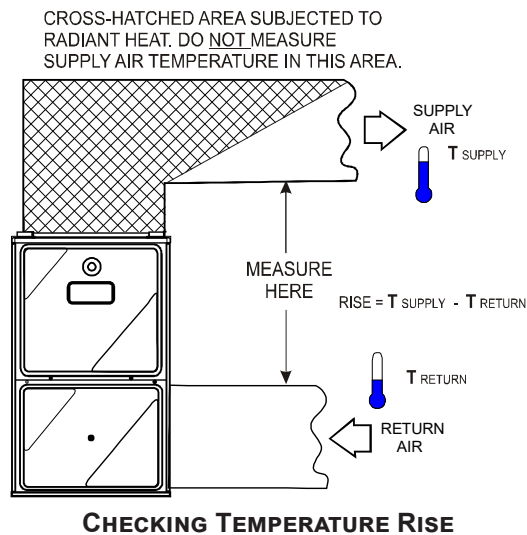
S-201 CHECKING TEMPERATURE RISE

The more air (CFM) being delivered through a given furnace, the less the rise will be; so the less air (CFM) being delivered, the greater the rise. The temperature rise should be adjusted in accordance to a given furnace specifications and its external static pressure. An incorrect temperature rise may result in condensing in or overheating of the heat exchanger. An airflow and temperature rise table is provided in the blower performance specification section. Determine and adjust temperature rise as follows:

1. Operate furnace with burners firing for approximately ten minutes. Check BTU input to furnace - do not exceed input rating stamped on rating plate. Ensure all registers are open and all duct dampers are in their final (fully or partially open) position.

SERVICING

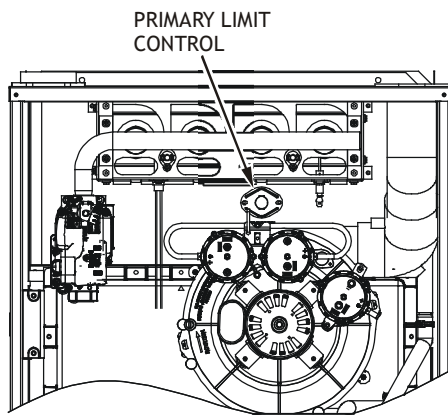
- Place thermometers in the return and supply ducts as close to the furnace as possible. Thermometers must not be influenced by radiant heat by being able to "see" the heat exchanger.



- Subtract the return air temperature from the supply air temperature to determine the air temperature rise. Allow adequate time for thermometer readings to stabilize.
- Adjust temperature rise by adjusting the circulator blower speed. Increase blower speed to reduce temperature rise. Decrease blower speed to increase temperature rise. Refer to Circulator Blower Speed section in the Product Design section of this manual for speed changing details. Temperature rise is related to the BTUH output of the furnace and the amount of air (CFM) circulated over the heat exchanger. Measure motor current draw to determine that the motor is not overloaded during adjustments.

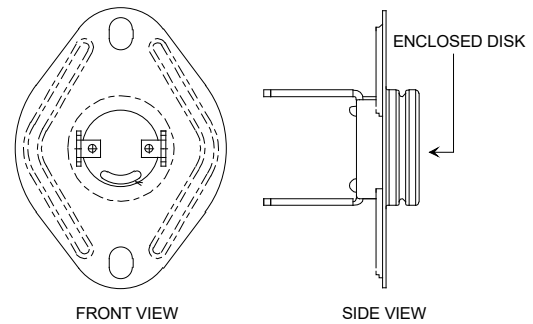
S-300 CHECKING PRIMARY LIMIT CONTROL

All primary limit controls are nonadjustable, automatic reset, bi-metal type limit control. Refer to the following drawing for the location of the primary limit.



**PRIMARY LIMIT CONTROL LOCATION
(90% UPFLOW FURNACE SHOWN, COUNTERFLOW
SIMILAR)**

The following drawing illustrates the style of limit switches used on the 90% furnaces.



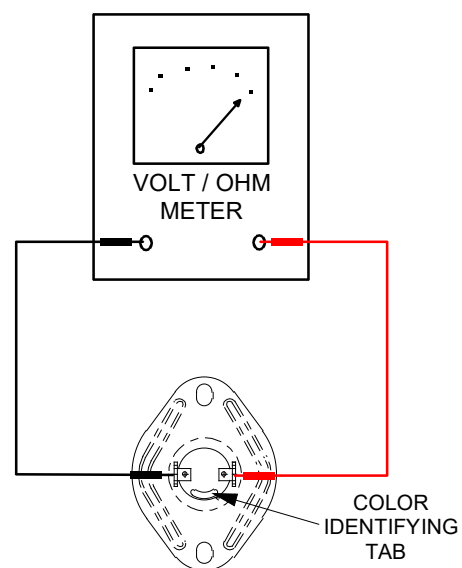
**PRIMARY LIMIT CONTROL STYLE
(90% FURNACES)**

WARNING

HIGH VOLTAGE

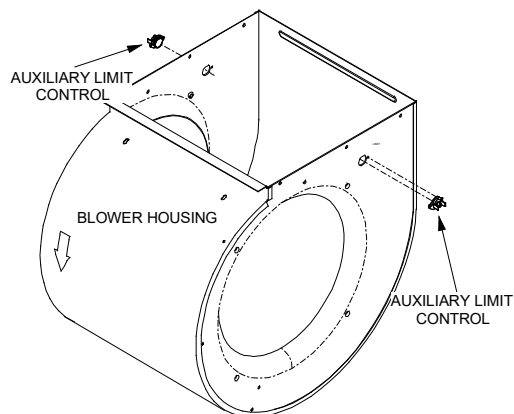
DISCONNECT **ALL** POWER BEFORE SERVICING OR CHANGING ANY ELECTRICAL WIRING. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.

- Remove burner compartment door to gain access to the primary limit.
- Remove low voltage wires at limit control terminals.
- With an ohmmeter, test between these two terminals as shown in the following drawing. The ohmmeter should read continuous unless heat exchanger temperature is above limit control setting. If not as above, replace the control.



**TESTING PRIMARY LIMIT CONTROL
(90% FURNACES)**

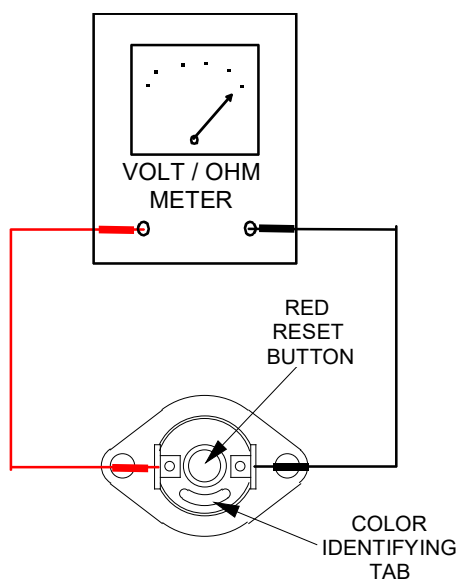
- After completing check and/or replacement of primary limit control, reinstall burner compartment door.
- Turn on electrical power and verify proper unit operation.

S-301 CHECKING AUXILIARY LIMIT CONTROL**AUXILIARY LIMIT CONTROL LOCATION****! WARNING****HIGH VOLTAGE**

DISCONNECT **ALL** POWER BEFORE SERVICING OR INSTALLING THIS UNIT. **MULTIPLE** POWER SOURCES MAY BE PRESENT. **FAILURE** TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.



1. Remove blower compartment door to gain access to the auxiliary.
2. Remove the wires from the auxiliary limit control terminals.
3. Using an ohmmeter, test for continuity across the two terminals.

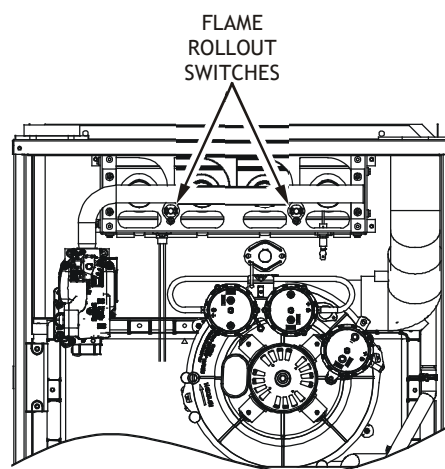
**TESTING AUXILIARY LIMIT CONTROL****! WARNING**

TO AVOID POSSIBLE FIRE, ONLY RESET THE AUXILIARY LIMIT CONTROL ONCE. IF IT SHOULD OPEN A SECOND TIME, A QUALIFIED SERVICER MUST DETERMINE WHY THE AUXILIARY LIMIT OPENED BEFORE RESETTING AGAIN.

NOTE: IF IT BECOMES NECESSARY TO SLIDE THE BLOWER ASSEMBLY OUT OF ANY 90% SINGLE-STAGE UPFLOW FURNACE, THE AUXILIARY LIMIT CONTROL MUST BE REMOVED FROM THE BLOWER DECK BEFORE THE BLOWER ASSEMBLY CAN BE REMOVED. AFTER THE BLOWER ASSEMBLY IS REINSTALLED, THE AUXILIARY LIMIT MUST BE REINSTALLED.

S-302 CHECKING FLAME ROLLOUT CONTROL

A temperature activated manual reset control is mounted to the manifold assembly as shown in the following illustration.

**FLAME ROLLOUT SWITCH LOCATION
(90% UPFLOW FURNACE SHOWN, COUNTERFLOW SIMILAR)**

The control is designed to open should a flame roll out occur. An over firing condition or flame impingement on the heat shield may also cause the control to open. If the rollout control opens, the air circulation blower will run continuously.

! WARNING

LINE VOLTAGE NOW PRESENT

1. Remove the burner compartment door to gain access to the rollout switch(es) mounted to burner bracket.
2. Reset the manual roll out switch
3. Remove wires from roll out switch
4. Using an ohmmeter, check for continuity across the switch.
5. If the switch will not close after manually resetting, it must be replaced.

SERVICING

6. Measure the voltage between each side of the rollout control and ground during the ignition attempt. If a rollout switch has tripped, it is important to find out why. Possible causes could be flame impingement, orifice plate out of position, burners with excessive cross-over slot dimension, over-firing, improper orifices, improper gas pressure, air leaking from around the heat exchanger into the burner compartment, air leaking through the heat exchanger itself.
7. After check and/or replacement of rollout switch, reinstall burner compartment door and verify proper unit operation.



WARNING

LINE VOLTAGE NOW PRESENT

1. Remove the burner compartment door to gain access to the rollout switch(es) mounted to burner bracket.
2. Reset the manual roll out switch
3. Remove wires from roll out switch
4. Using an ohmmeter, check for continuity across the switch.
5. If the switch will not close after manually resetting, it must be replaced.
6. Measure the voltage between each side of the rollout control and ground during the ignition attempt. If a rollout switch has tripped, it is important to find out why. Possible causes could be flame impingement, orifice plate out of position, burners with excessive cross-over slot dimension, over-firing, improper orifices, improper gas pressure, air leaking from around the heat exchanger into the burner compartment, air leaking through the heat exchanger itself.
7. After check and/or replacement of rollout switch, reinstall burner compartment door and verify proper unit operation.

S-303 INDUCED DRAFT BLOWER MOTOR



WARNING

HIGH VOLTAGE

DISCONNECT ALL POWER BEFORE SERVICING OR INSTALLING THIS UNIT. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.



1. Remove burner compartment door to gain access to the induced draft blower motor.
2. Disconnect the motor wire leads from its connection point at the induced draft motor.
3. Using a ohmmeter, test for continuity between each of the motor leads.

4. Touch one probe of the ohmmeter to the motor frame (ground) and the other probe in turn to each lead. If the windings do not test continuous or a reading is obtained to ground, replace the motor.
5. If the windings have a continuity reading, reconnect wires. Turn power on to the furnace and turn the thermostat on in the heating mode. Check voltage for 115V at the induced draft motor terminals during the trial for ignition. If you have 115V and the motor does not run, replace the induced draft motor.
6. After completing check and/or replacement of induced draft motor, reinstall burner compartment door.
7. Turn on electrical power and verify proper unit operation.

S-304 CHECKING GAS VALVE (REDUNDANT)

A combination redundant operator type gas valve which provides all manual and automatic control functions required for gas fired heating equipment is used.

The valve provides control of main burner gas flow, pressure regulation, and 100 percent safety shut-off.



WARNING

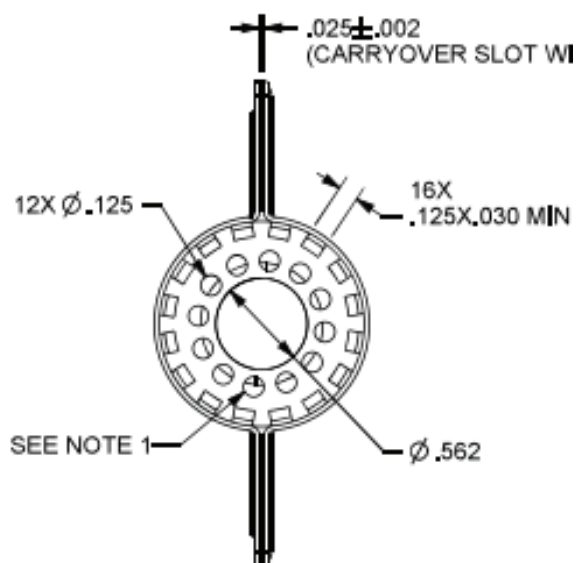
DISCONNECT ALL POWER BEFORE SERVICING

Single stage gas valves should be tested on the furnace with 24 VAC connected to the gas valve and manometers reading supply line and manifold pressures.

S-305 CHECKING MAIN BURNERS

Burners have been redesigned for 34.5" chassis furnaces. Overall length and width dimensions remain the same as 40" model burners. The burners used 34.5" models have burner head insert with larger diameter center hole and a larger number of surrounding holes.

The main burners are used to provide complete combustion of various fuels in a limited space, and transfer this heat of the burning process to the heat exchanger. Proper ignition, combustion, and extinction are primarily due to burner design, orifice sizing, gas pressure, primary and secondary air, vent and proper seating of burners.



BECKETT BURNER

**WARNING**

DISCONNECT ALL GAS AND ELECTRICAL POWER SUPPLY.

In checking main burners, look for signs of rust, oversized and undersized carry over ports restricted with foreign material, etc, burner cross-over slots should not be altered in size.

S-306 CHECKING ORIFICES

S model furnaces have factory installed #43 natural gas orifices.

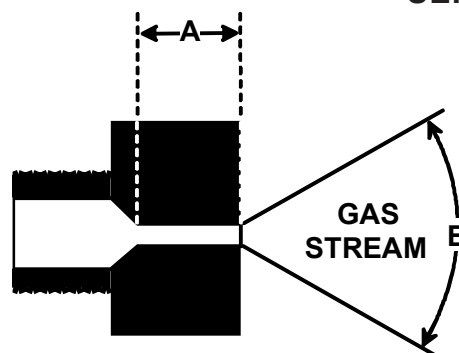
The only time resizing is required is when a reduction in firing rate is required for an increase in altitude or a furnace is being converted for use with LP gas.

Orifices should be treated with care in order to prevent damage. They should be removed and installed with a box-end wrench in order to prevent distortion. In no instance should an orifice be peened over and redrilled. This will change the angle or deflection of the vacuum effect or entraining of primary air, which will make it difficult to adjust the flame properly. This same problem can occur if an orifice spud of a different length is substituted.

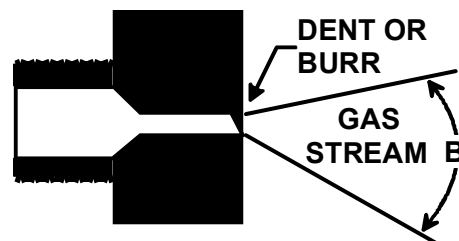
**WARNING**

DISCONNECT ALL GAS AND ELECTRICAL POWER SUPPLY.

1. Check orifice visually for distortion and/or burrs.
2. Check orifice size with orifice sizing drills.



The length of Dimension "A" determines the angle of Gas Stream "B".



A dent or burr will cause a severe deflection of the gas stream.

S-307 CHECKING GAS PRESSURE

Gas Supply Pressure Measurement

**CAUTION**

TO PREVENT UNRELIABLE OPERATION OR EQUIPMENT DAMAGE, THE INLET GAS SUPPLY PRESSURE MUST BE AS SPECIFIED ON THE UNIT RATING PLATE WITH ALL OTHER HOUSEHOLD GAS FIRED APPLIANCES OPERATING.

Gas inlet and manifold pressures should be checked and adjusted in accordance to the type of fuel being consumed.

The line pressure supplied to the gas valve must be within the range specified below. The supply pressure can be measured at the gas valve inlet pressure tap or at a hose fitting installed in the gas piping drip leg. The supply pressure must be measured with the burners operating. To measure the gas supply pressure, use the following procedure.

**WARNING**

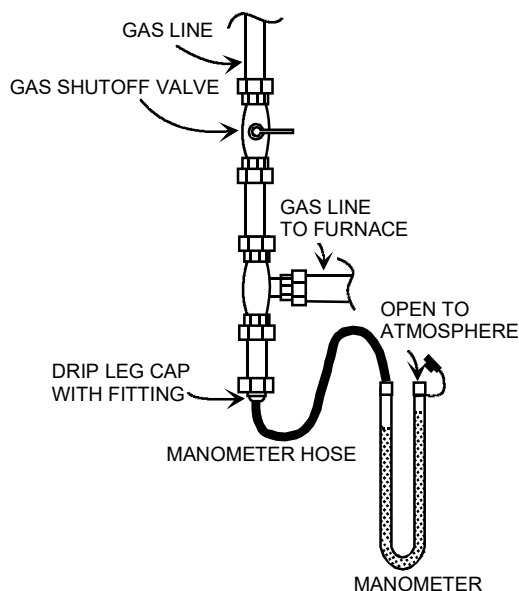
DISCONNECT ELECTRICAL POWER AND SHUT OFF GAS SUPPLY.

1. After turning off gas to furnace at the manual gas shutoff valve external to the furnace, remove burner compartment door to gain access to the gas valve.
2. Connect a calibrated water manometer (or appropriate gas pressure gauge) at either the gas valve inlet pressure tap or the gas piping drip leg as shown in the following figures.

SERVICING

NOTE: AT EITHER LOCATION, A HOSE FITTING MUST BE INSTALLED PRIOR TO MAKING THE HOSE CONNECTION.

NOTE: USE ADAPTER KIT #0151K00000S TO MEASURE GAS PRESSURE ON WHITE-RODGERS AND 36[G/J]54 GAS VALVES.



MEASURING INLET GAS PRESSURE (ALTERNATE METHOD)

- Turn ON the gas and electrical power supply and operate the furnace and all other gas consuming appliances on the same gas supply line.
- Measure furnace gas supply pressure with burners firing. Supply pressure must be within the range specified in the following table.

INLET GAS SUPPLY PRESSURE		
Natural Gas	Minimum: 4.5" w.c.	Maximum: 10.0" w.c.
Propane Gas	Minimum: 11.0" w.c.	Maximum: 13.0" w.c.

If supply pressure differs from above, make necessary adjustments to pressure regulator, gas piping size, etc., and/or consult with local gas utility.


WARNING



HIGH VOLTAGE
DISCONNECT **ALL** ELECTRICAL POWER AND SHUT OFF GAS SUPPLY BEFORE SERVICING OR INSTALLING THIS UNIT.
MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.

- Disconnect manometer after turning off gas at manual shutoff valve. Reinstall plug before turning on gas to furnace.
- Turn OFF any unnecessary gas appliances started in step 3.

- Turn on gas to furnace and check for leaks. If leaks are found, repair and then reinstall burner compartment door.
- Turn on electrical power and verify proper unit operation.


WARNING



HIGH VOLTAGE
DISCONNECT **ALL** ELECTRICAL POWER AND SHUT OFF GAS SUPPLY BEFORE SERVICING OR INSTALLING THIS UNIT.
MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.

GAS MANIFOLD PRESSURE MEASUREMENT AND ADJUSTMENT


CAUTION

TO PREVENT UNRELIABLE OPERATION OR EQUIPMENT DAMAGE, THE GAS MANIFOLD PRESSURE MUST BE AS SPECIFIED ON THE UNIT RATING PLATE. ONLY MINOR ADJUSTMENTS SHOULD BE MADE BY ADJUSTING THE GAS VALVE PRESSURE REGULATOR.

NOTE: USE ADAPTER KIT #0151K00000S TO MEASURE GAS PRESSURE ON WHITE-RODGERS 36J22 GAS VALVES.

Only small variations in gas pressure should be made by adjusting the gas valve pressure regulator. The manifold pressure must be measured with the burners operating. To measure and adjust the manifold pressure, use the following procedure.


WARNING

HIGH VOLTAGE
DISCONNECT **ALL** ELECTRICAL POWER AND SHUT OFF GAS SUPPLY BEFORE SERVICING OR INSTALLING.

- After turning off gas to furnace at the manual gas shutoff valve external to the furnace, remove burner compartment door to gain access to the gas valve.
- Connect a calibrated water manometer (or appropriate gas pressure gauge) at the gas valve outlet pressure tap. Refer to Measuring Gas Pressure: Single Stage Valves figure for single stage valve outlet pressure tap connections.


WARNING

LINE VOLTAGE NOW PRESENT

- Turn ON the gas and electrical power supply and operate the furnace.

SERVICING

4. Measure gas manifold pressure with burners firing. Adjust manifold pressure using the table below.

Manifold Gas Pressure	
Natural Gas	3.5" w.c.
Propane Gas	10.0" w.c.

The final manifold pressure must not vary more than ± 0.3 " w.c. from the above specified pressures. Any necessary major changes in gas flow rate should be made by changing the size of the burner orifice.

5. White-Rodgers 36J22 Valves:
- Back outlet pressure test screw (inlet/outlet pressure boss) out one turn (counterclockwise, not more than one turn).
 - Attach a hose and manometer to the outlet pressure outlet pressure boss.
 - Turn ON the gas supply.
 - Turn on power and close thermostat "R" and "W1" contacts to provide a call for heat.
 - Measure the gas manifold pressure with burners firing. Adjust manifold pressure using the Manifold Gas Pressure table shown below.
 - Remove regulator cover screw from the outlet pressure regulator adjust tower and turn screw clockwise to increase pressure or counterclockwise to decrease pressure. Replace regulator cover screw.
 - Turn off all electrical power and gas supply to the system.
 - Remove the manometer hose from the hose barb fitting or outlet pressure boss.
 - Turn outlet pressure test screw in to seal pressure port (clockwise, 7 in-lb minimum).
6. Honeywell VR8215 Valve:
- Remove the outlet pressure boss plug. Install an 1/8" NPT hose barb fitting into the outlet pressure tap.
 - Attach a hose and manometer to the outlet pressure barb fitting.
 - Turn ON the gas supply.
 - Turn on power and close thermostat "R" and "W1" contacts to provide a call for low stage heat.
 - Measure the gas manifold pressure with burners firing. Adjust manifold pressure using the Manifold Gas Pressure table shown below.
 - Remove regulator cover screw from the outlet pressure regulator adjust tower and turn screw clockwise to increase pressure or counterclockwise to decrease pressure. Replace regulator cover screw.
 - Turn off all electrical power and gas supply to the system.
 - Remove the manometer hose from the hose barb fitting or outlet pressure boss.

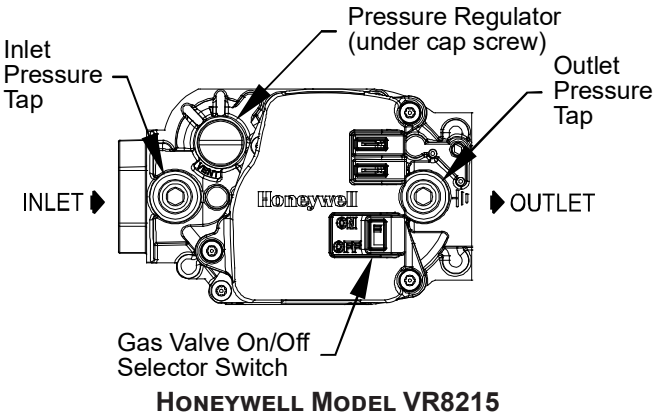
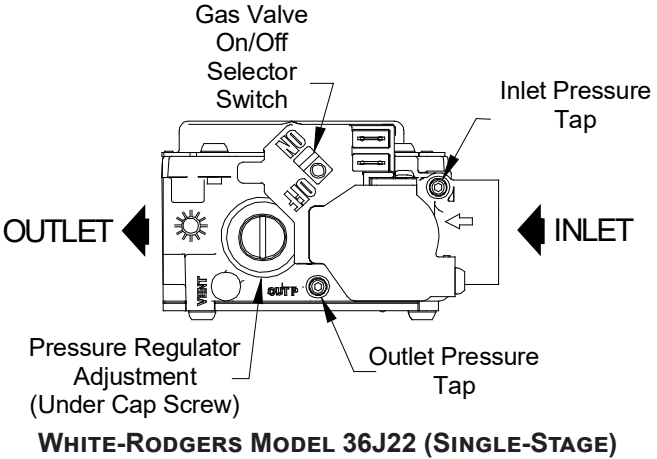
- i. Remove the 1/8" NPT hose barb fitting from the outlet pressure tap. Replace the outlet pressure boss plug and seal with a high quality thread sealer.

**WARNING**

HIGH VOLTAGE
DISCONNECT **ALL** ELECTRICAL POWER AND SHUT OFF GAS SUPPLY BEFORE SERVICING OR INSTALLING THIS UNIT.
MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.



7. Turn on gas to furnace and check for leaks. If leaks are found, repair and then reinstall burner compartment door.
8. Turn on electrical power and verify proper unit operation.



Manifold Gas Pressure			
Gas	Rate	Range	Nominal
Propane Gas	High Stage	9.7 to 10.3" w.c.	10.0" w.c.
	Low Stage	5.7 to 6.3" w.c.	6.0" w.c.

S-308 CHECKING HOT SURFACE IGNITOR

Single stage furnaces use a 115 volt silicon carbide igniter (p/n 0130F00008) with 17-second warm up time.

SERVICING



WARNING

DISCONNECT **ALL** POWER BEFORE SERVICING.

1. Remove burner compartment door to gain access to the ignitor.
2. Ignitor cool - approximately 70 - 77°F.
3. Disconnect the ignitor from the Ignition Control.
4. Using an ohmmeter measure the resistance of the ignitor: -at room temperature a normal reading will be 37 - 68 ohms.
5. Reconnect ignitor.



WARNING

LINE VOLTAGE NOW PRESENT

6. Place unit in heating cycle, measure current draw of ignitor during preheat cycle.
 - a. Turn off all electrical power and gas supply to the system.
 - b. Remove the manometer hose from the outlet pressure boss.
 - c. Remove the 1/8" NPT hose barb fitting from the outlet pressure tap. Replace the outlet pressure boss plug and seal with a high quality thread sealer.



WARNING

HIGH VOLTAGE

DISCONNECT **ALL** ELECTRICAL POWER AND SHUT OFF GAS SUPPLY BEFORE SERVICING OR INSTALLING.

7. Turn on gas to furnace and check for leaks. If leaks are found, repair and then reinstall burner compartment door.
8. Turn on electrical power and verify proper unit operation. Make sure furnace operates at the proper manifold pressure at both high and low stage outputs.

Manifold Gas Pressure

Gas	Rate	Range	Nominal
Natural Gas	High Stage	3.2 to 3.8" w.c.	3.5" w.c.
	Low Stage	1.6 to 2.2" w.c.	1.9" w.c.

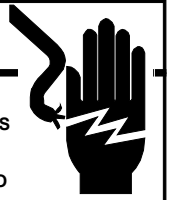


WARNING

HIGH VOLTAGE

DISCONNECT **ALL** ELECTRICAL POWER AND SHUT OFF GAS SUPPLY BEFORE SERVICING OR INSTALLING THIS UNIT.

MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.



WARNING

HIGH VOLTAGE

DISCONNECT **ALL** ELECTRICAL POWER AND SHUT OFF GAS SUPPLY BEFORE SERVICING OR INSTALLING.

Manifold Gas Pressure

Gas	Rate	Range	Nominal
Propane Gas	High Stage	9.7 to 10.3" w.c.	10.0" w.c.
	Low Stage	5.7 to 6.3" w.c.	6.0" w.c.

S-308A CHECKING HOT SURFACE IGNITOR

120V Silicon Nitride Igniter -Daikin Brand model furnaces use a 120V silicon nitride igniter for ignition. The normal operating temperature is approximately 2156°F - 2678°F. At room temperature the igniter ohm reading should be from 37-68 ohms.

1. Place unit in heating cycle, measure current draw of ignitor during preheat cycle.
The steady state current at 120V is 0.37 to 0.68 amps.
2. After checking and/or replacing of hot surface ignitor, reinstall burner compartment door and verify proper unit operation.

S-309 CHECKING FOR FLASHBACK

Flashback will also cause burning in the burner venturi, but is caused by the burning speed being greater than the gas-air flow velocity coming from a burner port.

Flashback may occur at the moment of ignition, after a burner heats up or when the burner turns off. The latter is known as extinction pop.

Since the end results of flashback and delayed ignition can be the same (burning in the burner venturi) a definite attempt should be made to determine which has occurred.

If flashback should occur, check for the following:

1. Improper gas pressure - adjust to proper pressure (See S-307 CHECKING GAS PRESSURE)..
2. Check burner for proper alignment and/or replace burner.
3. Improper orifice size - check orifice for obstruction.

S-310 CHECKING PRESSURE SWITCH

The pressure control is a safety device to prevent the combustion cycle from occurring with inadequate venting caused by a restricted or blocked vent pipe. In addition to the inducer pressure switch, S model furnaces have a "front cover pressure switch" wired in series with the gas valve. This pressure switch keeps the gas valve from opening in the event of condensate backing up in the secondary heat exchanger. This could occur from improperly connected drains or a plugged drain tube.


WARNING



HIGH VOLTAGE
 DISCONNECT ALL POWER BEFORE SERVICING OR INSTALLING THIS UNIT. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.

1. Remove burner compartment door to gain access to pressure switch(es).
2. Remove wires from the pressure switch(es) electrical terminals.
3. Remove the pressure control hose from the control and interconnect with an inclined manometer as shown in the following figures.
4. With an ohm meter connected across the pressure switch terminals; with the inducer running the switch should close and the ohm meter should show a complete circuit across the pressure switch. If the switch is not closed, compare the negative pressure to the closing point specified for the particular switch. Either the switch is defective or the inducer / venting system is inadequate.

S-311 HIGH ALTITUDE APPLICATION (USA)

A high altitude kit is required for installations above 7,000 ft. Refer to the accessory matrix in this manual to determine the proper natural gas and LP gas high altitude kit for your furnace.

S-312 CHECKING FOR DELAYED IGNITION

Delayed ignition is a delay in lighting a combustible mixture of gas and air which has accumulated in the combustion chamber.

Furnace design makes this extremely unlikely unless safety controls have been by-passed or tampered with. Never bypass or alter furnace controls.

If delayed ignition should occur, the following should be checked:

1. Improper gas pressure - adjust to proper pressure (See S-307 *Checking Gas Pressure*).
2. Improper burner positioning - burners should be in locating slots, level front to rear and left to right.
3. Carry over (lighter tube or cross lighter) obstructed - clean.

4. Main burner orifice(s) deformed, or out of alignment to burner - replace.

S-313 CHECKING INTEGRATED IGNITION CONTROL BOARDS

NOTE: FAILURE TO EARTH GROUND THE FURNACE, REVERSING THE NEUTRAL AND HOT WIRE CONNECTION TO THE LINE (POLARITY), OR A HIGH RESISTANCE CONNECTION IN THE NEUTRAL LINE MAY CAUSE THE CONTROL TO LOCKOUT DUE TO FAILURE TO SENSE FLAME.


WARNING

TO AVOID THE RISK OF ELECTRICAL SHOCK, WIRING TO THE UNIT MUST BE PROPERLY POLARIZED AND GROUNDED. DISCONNECT POWER BEFORE PERFORMING SERVICE LISTED BELOW.

The ground wire must run from the furnace all the way back to the electrical panel. Proper grounding can be confirmed by disconnecting the electrical power and measuring resistance between the neutral (white) connection and the burner closest to the flame sensor. Resistance should be less than 2 ohms.

The ignition control is a combination electronic and electromechanical device and is not field repairable. Complete unit must be replaced.


WARNING

LINE VOLTAGE NOW PRESENT

These tests must be completed within a given time frame due to the operation of the ignition control.

The ignition control is capable of diagnosing many furnace failures to help in troubleshooting. The trial for ignition period is 4 seconds.

1. Check for 120 volts from Line 1 (Hot) to Line 2 (Neutral) at the ignition control. No voltage, check the door switch connections and wire harness for continuity.
2. Check for 24 volts from W to C terminal on the ignition control. No voltage. Check transformer, room thermostat, and wiring.
If you have 24 volts coming off the transformer but receive approximately 13 volts on the terminal board between (C) and (R), check for blown fuse.
3. Check for 120 volts to the induced draft blower by measuring voltage between Pin 1 (on the 2-pin connector) and neutral located on circuit board. No voltage, check for loose connection in the 2-pin connector or replace ignition control.

- If voltage is present in Steps 1 through 3 and the induced draft blower is operating, check for 120 volts to the ignitor during the preheat cycle. Measure voltage between Pin 2 (on the 2-pin connector) and neutral located on ignition control. No voltage, check pressure switch or replace the ignition control board.
- After the ignitor warmup time, begin checking for 24 volts to the gas valve. Voltage will be present for four seconds only if proof of flame has been established.
- If proof of flame was established voltage will be provided to the air circulation blower following the heat on delay period.

- As soon as flame is established a micro-amp reading should be evident once proof of flame (micro-amp reading) is established, the hot surface ignitor will be de-energized.
- The Integrated Ignition controls will have 1 to 4 micro-amps. If the micro-amp reading is less than the minimum specified, check for high resistance wiring connections, sensor to burner gap, dirty flame sensor, or poor grounding.
- If absolutely no reading, check for continuity on all components and if good - replace ignition control module.

ERROR CODE	
Red LED Flash	Error / Condition
Flash Codes stored in memory (auto-erased after 14 days)	
1	System Lockout (Retries Exceeded)
2	Pressure Switch stuck Closed
3	Pressure Switch stuck Open
4	Open High Temperature Limit Switch
5	Flame Sensed>4.25 Seconds with Gas Valve De-Energized
6	Open High Temperature Limit Switch
8	Igniter Relay Fault
10	Open Fuse
11	Igniter Open
12	Inducer Relay Error
Flash Codes NOT stored in memory	
7	Low Flame Sense Signal
9	Twinning Fault
OFF	Control Failure / No Power / Internal Fault / IRQ Loss. Gas Heating Lockout (Gas valve energized when it should be de-energized) Or Gas Valve De-energized when it should be Energized
Rapid Flash	Reverse Polarity
Continuous On	Normal Operation

NOTE: CONTAMINATED FUEL OR COMBUSTION AIR CAN CREATE A NEARLY INVISIBLE COATING ON THE FLAME SENSOR. THIS COATING WORKS AS AN INSULATOR CAUSING A LOSS IN THE FLAME SENSE SIGNAL. IF THIS SITUATION OCCURS THE FLAME SENSOR MUST BE CLEANED WITH STEEL WOOL.

S-314 CHECKING FLAME SENSOR

A flame sensing device is used in conjunction with the ignition control module to prove combustion. If proof of flame is not present the control will de-energize the gas valve and “retry” for ignition or lockout.


WARNING



HIGH VOLTAGE

DISCONNECT **ALL** POWER BEFORE SERVICING OR INSTALLING THIS UNIT. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.

- Connect a micro-amp meter in series with this wire and the sensor terminal.
- Be sure the positive side of the meter is to Blue wire and the negative side of the meter is to sensor terminal.


WARNING

LINE VOLTAGE NOW PRESENT

- Place the unit into a heating cycle.

MAINTENANCE

 WARNING	
HIGH VOLTAGE DISCONNECT ALL POWER BEFORE SERVICING OR INSTALLING THIS UNIT. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.	

 CAUTION
IF YOU MUST HANDLE THE IGNITOR, HANDLE WITH CARE. TOUCHING THE IGNITOR BODY WITH BARE FINGERS, ROUGH HANDLING, OR VIBRATION COULD RESULT IN EARLY IGNITOR FAILURE. ONLY A QUALIFIED SERVICER SHOULD EVER HANDLE THE IGNITOR.

ANNUAL INSPECTION

The furnace should be inspected by a qualified installer, or service agency at least once per year. This check should be performed at the beginning of the heating season. This will ensure that all furnace components are in proper working order and that the heating system functions appropriately. Pay particular attention to the following items. Repair or service as necessary.

- Flue pipe system. Check for blockage and/or leakage. Check the outside termination and the connections at and internal to the furnace.
- Combustion air intake pipe system (where applicable). Check for blockage and/or leakage. Check the outside termination and the connection at the furnace.
- Heat exchanger. Check for corrosion and/or buildup within the heat exchanger passageways.
- Burners. Check for proper ignition, burner flame, and flame sense.
- Drainage system. Check for blockage and/or leakage. Check hose connections at and internal to furnace.
- Wiring. Check electrical connections for tightness and/or corrosion. Check wires for damage.
- Filters.

AIR FILTER

 WARNING
NEVER OPERATE FURNACE WITHOUT A FILTER INSTALLED AS DUST AND LINT WILL BUILD UP ON INTERNAL PARTS RESULTING IN LOSS OF EFFICIENCY, EQUIPMENT DAMAGE, AND POSSIBLE FIRE.

Filters must be used with this furnace. Filters do not ship with these furnaces but must be provided by the installer for proper furnace operation.

Remember that dirty filters are the most common cause of inadequate heating or cooling performance.

 WARNING	
HIGH VOLTAGE DISCONNECT ALL POWER BEFORE SERVICING, REMOVING THE FILTER OR PREFORMING ANY OTHER MAINTENANCE. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.	

MAINTENANCE

Improper filter maintenance is the most common cause of inadequate heating or cooling performance. Filters should be cleaned (permanent) or replaced (disposable) every two months or as required. It is the owner's responsibility to keep air filters clean. When replacing a filter, it must be replaced with a filter of the same type and size.

FILTER REMOVAL

Depending on the installation, differing filter arrangements can be applied. Filters can be installed in the central return register, the bottom of the blower compartment (upflow only), a side panel external filter rack kit (upflow only), or the ductwork above a counterflow furnace. A media air filter or electronic air cleaner can be used as an alternate filter. The filter sizes given in the Product Design section of this manual or the product Specification Sheet must be followed to ensure proper unit performance. Refer to the following information for removal and installation of filters.

FILTER REMOVAL PROCEDURE

MEDIA AIR FILTER OR ELECTRONIC AIR CLEANER REMOVAL

Follow the manufacturer's directions for service.

UPRIGHT COUNTERFLOW FILTER REMOVAL

To remove filters from the ductwork above an upright counterflow installation:

1. Turn off electrical power to furnace.
2. Remove access door in ductwork above furnace.
3. Remove filters.
4. Remove blower compartment door. Vacuum compartment. Replace blower compartment door.
5. Replace filters opposite of removal.
6. Replace access door in ductwork.

HORIZONTAL UNIT FILTER REMOVAL

Filters in horizontal installations are located in the central return register.

INDUCED DRAFT AND CIRCULATION BLOWERS

The bearings in the induced draft blower and circulator blower motors are permanently lubricated by the manufacturer. No further lubrication is required. Check motor windings for accumulation of dust which may cause overheating. Clean as necessary.

CONDENSATE DRAINAGE SYSTEM (QUALIFIED SERVICER ONLY)

The drain tubes, standpipe, and field supplied drain line must be checked annually and cleaned as often as necessary to ensure proper condensate drainage.

FLAME SENSOR (QUALIFIED SERVICER ONLY)

Under some conditions, the fuel or air supply can create a nearly invisible coating on the flame sensor. This coating acts as an insulator, causing a drop in the flame sensing signal. If this occurs, a qualified servicer must carefully clean the flame sensor with steel wool. After cleaning, the flame sensor output should be as listed on the specification sheet.

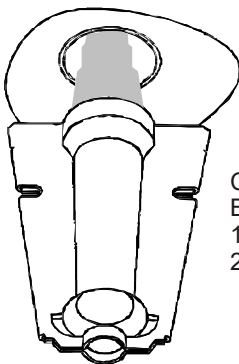
BURNERS


WARNING



HIGH VOLTAGE
ELECTRICAL COMPONENTS ARE CONTAINED IN BOTH COMPARTMENTS. TO AVOID ELECTRICAL SHOCK, INJURY OR DEATH, DO NOT REMOVE ANY INTERNAL COMPARTMENT COVERS OR ATTEMPT ANY ADJUSTMENT. CONTACT A QUALIFIED SERVICE AGENT AT ONCE IF AN ABNORMAL FLAME APPEARANCE SHOULD DEVELOP.

Periodically during the heating season make a visual check of the burner flames. Turn the furnace on at the thermostat. Wait a few minutes since any dislodged dust will alter the normal flame appearance. Flames should be stable, quiet, soft and blue with slightly orange tips. They should not be yellow. They should extend directly outward from the burner ports without curling downward, floating or lifting off the ports.



Check the Burner Flames for:

1. Stable, soft and blue.
2. Not curling, floating or lifting off.

BURNER FLAME

TEST EQUIPMENT

Proper test equipment for accurate diagnosis is as essential as regular hand tools.

The following is a must for every service technician and service shop.

1. Dial type thermometers or thermocouple meter (optional) - to measure dry bulb temperature.
2. Amprobe - to measure amperage and voltage.
3. Volt-Ohm Meter - testing continuity, capacitors, and motor windings.
4. Inclined Manometer - to measure static pressure, pressure drop across coils, filters, and draft.
5. Water Manometer (12") - to test gas inlet and manifold pressure.

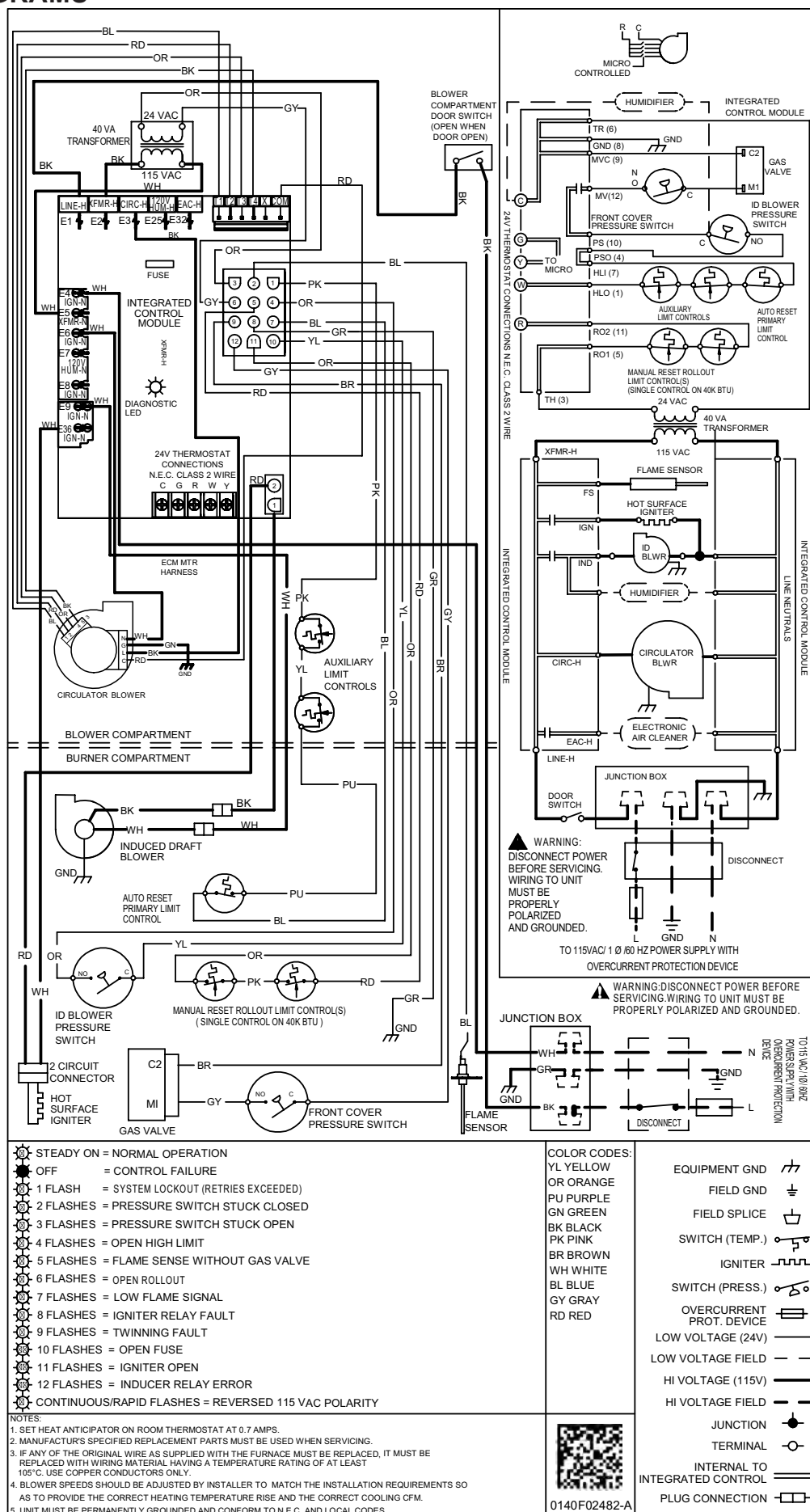
Other recording type instruments can be essential in solving abnormal problems, however, in many instances they may be rented from local sources.

SERVICING

MODEL	PRIMARY LIMIT SETTING	PRIMARY LIMIT PART NO.	AUX LIMIT SETTING	AUX LIMIT PART NO.	R.O.S LIMIT SETTING	R.O.S. LIMIT PART NO.
DM96SE0403ANAA	170 ±5	20162906	120	0130F00038	230 ±6	10123538
DM96SE0603BNAA	160 ±5	20162903	120	0130F00038	240 ±10	10123539
DM96SE0803BNAA	150 ±5	20162904	120	0130F00038	230 ±6	10123538
DM96SE0804CNAA	160 ±5	20162903	120	0130F00038	240 ±10	10123539
DM96SE0805CNAA	160 ±5	20162903	120	0130F00038	250 ±9	0130F00123
DM96SE1005CNAA	150 ±5	20162904	120	0130F00038	230 ±6	10123538
DM96SE1205DNAA	150 ±5	20162904	120	0130F00038	275 ±7	10123528
DC96SE0403BNAA	210 ±6	20162901	120	0130F00038	250 ±9	0130F00123
DC96SE0603BNAA	210 ±6	20162901	120	0130F00038	250 ±9	0130F00123
DC96SE0804CNAA	200 ±5	0130F00071	120	0130F00038	220 ±7	10123534
DC96SE1005CNAA	160 ±5	20162903	120	0130F00038	240 ±10	10123539
DC96SE1205DNAA	170 ±5	20162906	120	0130F00038	240 ±10	10123539

DM96SE / DC96SE Pressure Switch Trip Point And Usage Chart					
MODEL	Coil Cover Set Point on Pressure Fall W. C.	Coile Cover Max Make Pressure W.C.	ID Blower Set Point on Pressure Fall W. C.	ID Blower Max Make Pressure W. C.	ID Blower Coil Cover Pressure Switch Assembly Part #
DM96SE0403ANAA	- 0.10 ± .05	- 0.25	- 0.35 ± .05	- 0.50	0130F00642
DM96SE0603BNAA	- 0.10 ± .05	- 0.25	- 1.49 ± .07	- 1.70	0130F00478
DM96SE0803BNAA	- 0.10 ± .05	- 0.25	- 1.27 ± .07	- 1.48	0130F00477
DM96SE0804CNAA	- 0.10 ± .05	- 0.25	- 1.27 ± .07	- 1.48	0130F00477
DM96SE0805CNAA	- 0.10 ± .05	- 0.25	- 1.20 ± .05	- 1.41	0130F00480
DM96SE1005CNAA	- 0.10 ± .05	- 0.25	- 0.97 ± .05	- 1.12	0130F00479
DM96SE1205DNAA	- 0.10 ± .05	- 0.25	- 0.75 ± .07	- 0.92	0130F00469
DC96SE0403BNAA	- 0.08 ± .04	- 0.16	- 1.49 ± .07	- 1.70	0130F00484
DC96SE0603BNAA	- 0.10 ± .05	- 0.25	- 1.27 ± .07	- 1.48	0130F00477
DC96SE0804CNAA	- 0.10 ± .05	- 0.25	- 1.27 ± .07	- 1.48	0130F00477
DC96SE1005CNAA	- 0.10 ± .05	- 0.25	- 1.27 ± .07	- 1.48	0130F00477
DC96SE1205DNAA	- 0.10 ± .05	- 0.25	- 0.85 ± .05	- 1.00	0130F00476

WIRING DIAGRAMS



Wiring is subject to change. Always refer to the wiring diagram on the unit for the most up-to-date wiring.